Mathematical Analysis

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Feel free to open issues or pull requests for corrections.

Contents

Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

  $$(P \vee Q) \wedge R$$

  is evaluated by first computing $P \vee Q$.

- The **logical connectives** are interpreted in the order

  $$\neg \ > \ \wedge \ > \ \oplus \ > \ \vee \ > \ \Longrightarrow, \ \Longleftarrow \ > \ \Longleftrightarrow$$

  Hence

  $$P \oplus Q \implies \neg R \vee S \wedge T \iff U \impliedby V$$

  is understood as

  $$((P \oplus Q) \implies ((\neg R) \vee (S \wedge T))) \iff (U \impliedby V)$$

- A **quantifier** binds the entire formula that follows it. Thus

  $$\forall x, P(x) \wedge Q(x) \implies R(x)$$

  means

  $$\forall x\, ((P(x) \wedge Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

  $$\begin{gathered} P \\ \Downarrow \\ Q \end{gathered}$$

  merely expresses that P is true, and therefore Q is true in the current discussion; it is not itself a proposition.

Chapter 1

The Number Systems

1.1 The Natural Number System

Axiom 1.1.1. The set of **natural numbers** $\mathbb{N}$ satisfies the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- $\forall S \subseteq \mathbb{N}, (1 \in S \land (\forall n \in S, s(n) \in S)) \implies S = \mathbb{N}$

$s(n)$ is called the **successor** of n. These axioms are called the **Peano axioms**.

Remark. The **Peano axioms** do not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** are not discussed, so we will just assume that $\mathbb{N}$ exists.

Theorem 1.1.2. *Suppose $P(n)$ is a statement about $n \in \mathbb{N}$. Define*

$$S := \{n \in \mathbb{N} \mid P(n) \textit{ is true}\} \subseteq \mathbb{N}$$

Suppose

- *$P(1)$ is true*
- *$\forall n \in \mathbb{N}, P(n)$ is true $\implies P(s(n))$ is true*

Then

- $1 \in S$
- $\forall n \in S, s(n) \in S$

By axiom 1.1.1,

$$S = \mathbb{N}$$
$$\Downarrow$$
$$\forall n \in \mathbb{N}, P(n) \textit{ is true}$$

*This method of proof is called the **principle of mathematical induction**.*

Definition 1.1.3. Suppose $n, m \in \mathbb{N}$.

- The **addition** $+$ on $\mathbb{N}$ is defined by
  - $n + 1 := s(n)$
  - $n + s(m) := s(n + m)$
- The **multiplication** $\cdot$ on $\mathbb{N}$ is defined by
  - $n \cdot 1 := n$
  - $n \cdot s(m) := n \cdot m + n$

Remark. Strictly speaking, definition 1.1.3 requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operations on $\mathbb{N}$ are **well-defined**.

Theorem 1.1.4. $\mathbb{N}$ *has the **well-ordering property**, which states that*

$$\forall S \subseteq \mathbb{N}, S \neq \varnothing \implies \exists m \in S, m = \min S$$

Proof. Assume

$$\exists S \subseteq \mathbb{N}, S \neq \varnothing \wedge (\nexists m \in S, m = \min S)$$

Define

- $T := \mathbb{N} \setminus S \subseteq \mathbb{N}$
- $L := \{n \in \mathbb{N} \mid \{1, 2, \ldots, n\} \subseteq T\} \subseteq \mathbb{N}$

We will show

$$L = \mathbb{N}$$
$$\Downarrow$$
$$T = \mathbb{N}$$
$$\Downarrow$$
$$S = \varnothing$$
$$\Downarrow$$
$$\perp$$

- **Base case:** Assume $1 \in S$. Then

$$\begin{gathered} 1 = \min S \\ \Downarrow \\ \bot \end{gathered}$$

  Therefore,

$$\begin{gathered} 1 \notin S \\ \Downarrow \\ 1 \in T \\ \Downarrow \\ 1 \in L \end{gathered}$$

- **Inductive step:** Suppose $n \in L$. Then

$$\begin{gathered} n \in L \\ \Downarrow \\ \{1, 2, \ldots, n\} \subseteq T \\ \Downarrow \\ \{1, 2, \ldots, n\} \cap S = \varnothing \end{gathered}$$

  Assume $n + 1 \in S$. Then

$$\begin{gathered} n + 1 = \min S \\ \Downarrow \\ \bot \end{gathered}$$

  Therefore,

$$\begin{gathered} n + 1 \notin S \\ \Downarrow \\ n + 1 \in T \\ \Downarrow \\ \{1, 2, \ldots, n + 1\} \subseteq T \\ \Downarrow \\ n + 1 \in L \end{gathered}$$

By theorem 1.1.2,

$$L = \mathbb{N}$$

□

1.2 Relations

Definition 1.2.1. A **relation** R from A to B is defined by

$$R \subseteq A \times B$$

In particular, a relation R on A is defined by

$$R \subseteq A \times A$$

The notation xRy denotes $(x, y) \in R$.

Definition 1.2.2. A relation $\sim$ on S is called an **equivalence relation** if it satisfies

- **Reflexivity**: $\forall x \in S, x \sim x$
- **Symmetry**: $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity**: $\forall x, y, z \in S, x \sim y \wedge y \sim z \implies x \sim z$

Definition 1.2.3. A collection $\{A_\alpha\}_{\alpha \in I}$ is called a **partition** of S if

- $\forall \alpha \in I, A_\alpha \neq \varnothing \wedge A_\alpha \subseteq S$
- $\forall \alpha, \beta \in I, \alpha \neq \beta \implies A_\alpha \cap A_\beta = \varnothing$
- $\displaystyle\bigcup_{\alpha \in I} A_\alpha = S$

Definition 1.2.4. Suppose $\sim$ is an equivalence relation on S. The **equivalence class** of $x \in S$ is defined by

$$[x] \coloneqq \{y \in S \mid y \sim x\}$$

The set of all equivalence classes of S is denoted by

$$S/\sim \coloneqq \{[x] \mid x \in S\}$$

Lemma 1.2.5. *Suppose $\sim$ is an equivalence relation on S. Then*

$$\forall x, y \in S, [x] \cap [y] \neq \varnothing \implies [x] = [y]$$

Proof. Suppose

- $x, y \in S$
- $[x] \cap [y] \neq \varnothing$

Then

$$\begin{gathered}\exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y\end{gathered}$$

Therefore,

$$\begin{gathered}\forall w \in S, w \sim x \implies w \sim y \\ \Downarrow \\ \forall w \in [x], w \in [y] \\ \Downarrow \\ [x] \subseteq [y]\end{gathered}$$

Similarly,

$$[y] \subseteq [x]$$

□

Theorem 1.2.6. *Suppose $\sim$ is an equivalence relation on S. Then*

$$\{[\alpha]\}_{[\alpha]\in S/\sim} = S/\sim$$

is a partition of S.

Proof. By definition 1.2.3,

- Suppose $[\alpha] \in S/\sim$. Then

  $$[\alpha] \subseteq S$$

  Also,

  $$\begin{gathered}\alpha \in [\alpha] \\ \Downarrow \\ [\alpha] \neq \varnothing\end{gathered}$$

- Suppose $[\alpha], [\beta] \in S/\sim$. By lemma 1.2.5,

  $$[\alpha] \neq [\beta] \implies [\alpha] \cap [\beta] = \varnothing$$

- – ($\subseteq$)

  $$\begin{gathered}\forall [\alpha] \in S/\sim, [\alpha] \subseteq S \\ \Downarrow \\ \bigcup_{[\alpha]\in S/\sim} [\alpha] \subseteq S\end{gathered}$$

– ($\supseteq$) Suppose $\beta \in S$. Then

$$\begin{gathered}[\beta] \in S/\sim \\ \Downarrow \\ \beta \in [\beta] \subseteq \bigcup_{[\alpha]\in S/\sim} [\alpha]\end{gathered}$$

□

Theorem 1.2.7. *Suppose $\{A_\alpha\}_{\alpha\in I}$ is a partition of S. Define $\sim$ on S by*

$$\forall x, y \in S, x \sim y \iff \exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$$

Then

(A) *$\sim$ is an equivalence relation on S*

(B) *$\{A_\alpha\}_{\alpha\in I} = S/\sim$*

Proof.

(A) By definition 1.2.2,

- **Reflexivity**: Suppose $x \in S$. Then

$$\begin{gathered}x \in \bigcup_{\alpha\in I} A_\alpha \\ \Downarrow \\ \exists \alpha \in I, x \in A_\alpha \\ \Downarrow \\ x \sim x\end{gathered}$$

- **Symmetry**: Suppose
  - $x, y \in S$
  - $x \sim y$

  Then

$$\begin{gathered}\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha \\ \Downarrow \\ y \in A_\alpha \wedge x \in A_\alpha \\ \Downarrow \\ y \sim x\end{gathered}$$

- **Transitivity**: Suppose
  - $x, y, z \in S$

  - $x \sim y$
  - $y \sim z$

  Then

  - $\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$
  - $\exists \beta \in I, y \in A_\beta \wedge z \in A_\beta$

  Therefore,

$$\begin{gathered} y \in A_\alpha \wedge y \in A_\beta \\ \Downarrow \\ A_\alpha \cap A_\beta \neq \varnothing \\ \Downarrow \\ \alpha = \beta \\ \Downarrow \\ x \in A_\alpha \wedge z \in A_\alpha \\ \Downarrow \\ x \sim z \end{gathered}$$

(B) Suppose $x \in S$. Then

$$\begin{gathered} x \in \bigcup_{\alpha \in I} A_\alpha \\ \Downarrow \\ \exists \alpha \in I, x \in A_\alpha \end{gathered}$$

First, we will show $A_\alpha = [x]$.

- ($\subseteq$) Suppose $y \in A_\alpha$. Then

$$\begin{gathered} y \in A_\alpha \wedge x \in A_\alpha \\ \Downarrow \\ y \sim x \\ \Downarrow \\ y \in [x] \end{gathered}$$

- ($\supseteq$) Suppose $y \in [x]$. Then

$$\begin{gathered} y \sim x \\ \Downarrow \\ \exists \beta \in I, y \in A_\beta \wedge x \in A_\beta \\ \Downarrow \\ x \in A_\alpha \wedge x \in A_\beta \\ \Downarrow \\ A_\alpha \cap A_\beta \neq \varnothing \\ \Downarrow \\ \alpha = \beta \\ \Downarrow \\ y \in A_\alpha \end{gathered}$$

Next, we will show $\{A_\alpha\}_{\alpha\in I} = S/\sim$.

- ($\subseteq$)

$$\begin{gathered}\forall \alpha \in I, A_\alpha \neq \varnothing \\ \Downarrow \\ \exists x \in A_\alpha \subseteq S \\ \Downarrow \\ A_\alpha = [x] \\ \Downarrow \\ \{A_\alpha\}_{\alpha\in I} \subseteq S/\sim\end{gathered}$$

- ($\supseteq$)

$$\begin{gathered}\forall x \in S, \exists \alpha \in I, A_\alpha = [x] \\ \Downarrow \\ S/R \subseteq \{A_\alpha\}_{\alpha\in I}\end{gathered}$$

□

1.3 The Integer and Rational Number Systems

Theorem 1.3.1. *Define $\sim$ on $\mathbb{N}\times\mathbb{N}$ by*

$$\forall (a,b),(c,d) \in \mathbb{N}\times\mathbb{N}, (a,b)\sim(c,d) \iff a+d=b+c$$

Then $\sim$ is an equivalence relation on $\mathbb{N}\times\mathbb{N}$.

Proof. By definition 1.2.2,

- **Reflexivity**: Suppose $(a,b)\in\mathbb{N}\times\mathbb{N}$. Then

$$\begin{gathered}a\in\mathbb{N}\wedge b\in\mathbb{N} \\ \Downarrow \\ a+b=b+a \\ \Downarrow \\ (a,b)\sim(a,b)\end{gathered}$$

- **Symmetry**: Suppose
  - $(a,b),(c,d)\in\mathbb{N}\times\mathbb{N}$
  - $(a,b)\sim(c,d)$

  Then

$$\begin{gathered}a+d=b+c \\ \Downarrow \\ c+b=d+a \\ \Downarrow \\ (c,d)\sim(a,b)\end{gathered}$$

- **Transitivity**: Suppose
  - $(a, b), (c, d), (e, f) \in \mathbb{N} \times \mathbb{N}$
  - $(a, b) \sim (c, d)$
  - $(c, d) \sim (e, f)$

  Then
$$\begin{gathered} a + d = b + c \wedge c + f = d + e \\ \Downarrow \\ a + d + f = b + c + f = b + d + e \\ \Downarrow \\ a + f = b + e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

$\square$

Remark. Strictly speaking, theorem 1.3.1 requires several **properties** of operations on $\mathbb{N}$. In this textbook, the properties are not discussed, so just **accepted** as true. The same issue arises for theorem 1.3.4.

Definition 1.3.2. Recall the relation $\sim$ defined in theorem 1.3.1. The set of **integers** $\mathbb{Z}$ is defined by

$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim$$

Definition 1.3.3. Suppose $[(a, b)], [(c, d)] \in \mathbb{Z}$.

- The **addition** $+$ on $\mathbb{Z}$ is defined by

$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

- The **multiplication** $\cdot$ on $\mathbb{Z}$ is defined by

$$[(a, b)] \cdot [(c, d)] := [(a \cdot c + b \cdot d, a \cdot d + b \cdot c)]$$

Remark. The operations on $\mathbb{Z}$ are **well-defined**; that is, they do not depend on the **choice of representatives** of the equivalence classes. Indeed, if

- $(a, b) \sim (a', b')$
- $(c, d) \sim (c', d')$

then one can verify that

- $(a+c, b+d) \sim (a'+c', b'+d')$
- $(a \cdot c + b \cdot d, a \cdot d + b \cdot c) \sim (a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c')$

The same issue arises for operations on $\mathbb{Q}$, whose **well-definedness** could be verified in exactly the same manner.

Theorem 1.3.4. *Define*

$$\mathbb{Z}^+ := \{[(n+1, 1)] \in \mathbb{Z} \mid n \in \mathbb{N}\} \subseteq \mathbb{Z}$$

Define $\sim$ *on* $\mathbb{Z} \times \mathbb{Z}^+$ *by*

$$\forall (a,b), (c,d) \in \mathbb{Z} \times \mathbb{Z}^+, (a,b) \sim (c,d) \iff a \cdot d = b \cdot c$$

Then $\sim$ *is an equivalence relation on* $\mathbb{Z} \times \mathbb{Z}^+$.

Proof. By definition 1.2.2,

- **Reflexivity**: Suppose $(a,b) \in \mathbb{Z} \times \mathbb{Z}^+$. Then

$$\begin{gathered} a \in \mathbb{Z} \wedge b \in \mathbb{Z}^+ \\ \Downarrow \\ a \cdot b = b \cdot a \\ \Downarrow \\ (a,b) \sim (a,b) \end{gathered}$$

- **Symmetry**: Suppose
  - $(a,b), (c,d) \in \mathbb{Z} \times \mathbb{Z}^+$
  - $(a,b) \sim (c,d)$

  Then

$$\begin{gathered} a \cdot d = b \cdot c \\ \Downarrow \\ c \cdot b = d \cdot a \\ \Downarrow \\ (c,d) \sim (a,b) \end{gathered}$$

- **Transitivity**: Suppose
  - $(a,b), (c,d), (e,f) \in \mathbb{Z} \times \mathbb{Z}^+$
  - $(a,b) \sim (c,d)$
  - $(c,d) \sim (e,f)$

Then

$$\begin{gathered} a \cdot d = b \cdot c \wedge c \cdot f = d \cdot e \\ \Downarrow \\ a \cdot d \cdot f = b \cdot c \cdot f = b \cdot d \cdot e \\ \Downarrow \\ a \cdot f = b \cdot e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

$\square$

Definition 1.3.5. Recall the relation $\sim$ defined in theorem 1.3.4. The set of **rational numbers** $\mathbb{Q}$ is defined by

$$\mathbb{Q} := (\mathbb{Z} \times \mathbb{Z}^+) / \sim$$

Definition 1.3.6. Suppose $[(a, b)], [(c, d)] \in \mathbb{Q}$.

- The **addition** $+$ on $\mathbb{Q}$ is defined by

$$[(a, b)] + [(c, d)] := [(ad + bc, bd)]$$

- The **multiplication** $\cdot$ on $\mathbb{Q}$ is defined by

$$[(a, b)] \cdot [(c, d)] := [(ac, bd)]$$

Remark. Strictly speaking, the constructions of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ produce **different kind** of sets. For example,

- $n \in \mathbb{N}$
- $[(n + 1, 1)] \in \mathbb{Z}$
- $[(n, [(2, 1)])] \in \mathbb{Q}$

are distinct objects in the underlying set-theoretic construction. Nevertheless, there are **canonical embeddings**

- $\mathbb{N} \to \mathbb{Z}$ such that $n \mapsto [(n + 1, 1)]$
- $\mathbb{Z} \to \mathbb{Q}$ such that $a \mapsto [(a, [(2, 1)])]$

which preserve the **algebraic operations**. Therefore, we identify each number system with its image in the next larger one and write

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$$

This identification is justified by the fact that the embeddings are canonical, requiring **no arbitrary choices**. Throughout this textbook, inclusions such as

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

should be understood as canonical identifications rather than literal inclusions arising directly from the set-theoretic constructions.

Remark. Define

$$0 \coloneqq [(1,1)] \in \mathbb{Z}$$

The notations below denote subsets of $\mathbb{Z}$:

- $[0,\infty)_{\mathbb{N}} \coloneqq \mathbb{N} \cup \{0\} \subseteq \mathbb{Z}$
- $\forall n \in [0,\infty)_{\mathbb{N}}, [n,\infty)_{\mathbb{N}} \coloneqq \{k \in [0,\infty)_{\mathbb{N}} \mid k \geq n\}$
- $\forall n, m \in [0,\infty)_{\mathbb{N}}, [n,m]_{\mathbb{N}} \coloneqq \{k \in [n,\infty)_{\mathbb{N}} \mid k \leq m\}$

1.4 Ordered Sets

Definition 1.4.1. Suppose a relation $<$ on S satisfies the following axioms:

- $\forall x, y, z \in S, x < y \wedge y < z \implies x < z$
- $\forall x, y \in S, x < y \oplus x = y \oplus y < x$

Then

- The relation $<$ is called an **(total) order** on S
- The structure $(S, <)$ is called an **(totally) ordered set**

The notation

- $x > y$ denotes $y < x$
- $x \geq y$ denotes $x > y \vee x = y$
- $x \leq y$ denotes $y \geq x$

Remark. An ordered set $(S,<)$ is a **structure**, distinct from its **underlying set** S. For any structure $\mathcal{S}$, the notation $|\mathcal{S}|$ denotes its underlying set, not its **cardinality**. That is,

$$S = |(S,<)| \neq (S,<)$$

The same issue arises for sections 1.5 and 2.4.

Remark. $(\mathbb{N},<), (\mathbb{Z},<), (\mathbb{Q},<)$ are ordered sets with

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N}, n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+, n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+, p = q + r$

where

$$\mathbb{Q}^+ := \left\{ [(p,q)] \in \mathbb{Q} \;\middle|\; p, q \in \mathbb{Z}^+ \right\} \subseteq \mathbb{Q}$$

Throughout this textbook, the **number systems** use the **standard order relations** above, unless otherwise specified.

Definition 1.4.2. Suppose $E \subseteq |(S,<)|$. Then

- E is said to be **bounded below** if

  $$\exists m \in S, \forall x \in E, m \leq x$$

  Such an m is called a **lower bound** of E.

- E is said to be **bounded above** if

  $$\exists M \in S, \forall x \in E, x \leq M$$

  Such an M is called an **upper bound** of E.

Also,

- The set of all lower bounds of E is denoted by

  $$E^{\downarrow} := \{m \in S \mid \forall x \in E, m \leq x\}$$

- The set of all upper bounds of E is denoted by

  $$E^{\uparrow} := \{M \in S \mid \forall x \in E, x \leq M\}$$

Definition 1.4.3. Suppose $E \subseteq |(S,<)|$. Then

- An element $m \in E$ is called a **minimum** of E if
$$m \in E^{\downarrow}$$
Such an m is denoted by
$$m = \min E$$
- An element $M \in E$ is called a **maximum** of E if
$$M \in E^{\uparrow}$$
Such an M is denoted by
$$M = \max E$$

Theorem 1.4.4. *Suppose*

- $E \subseteq \mathbb{Z}$
- $E \neq \varnothing$

Then

(A) $E^{\downarrow} \neq \varnothing \implies \exists m \in E, m = \min E$

(B) $E^{\uparrow} \neq \varnothing \implies \exists M \in E, M = \max E$

Proof.

(A) Suppose $E^{\downarrow} \neq \varnothing$. Then
$$\exists k \in \mathbb{Z}, \forall \alpha \in E, k \leq \alpha$$
Define
$$F := \{\alpha - k + 1 \mid \alpha \in E\} \subseteq \mathbb{N}$$
Then
$$\begin{gathered} F \neq \varnothing \\ \Downarrow \qquad \text{Theorem 1.1.4} \\ \exists n \in F, n = \min F \\ \Downarrow \\ \forall \alpha \in E, n \leq \alpha - k + 1 \\ \Downarrow \\ \forall \alpha \in E, n + k - 1 \leq \alpha \end{gathered}$$
Therefore,
$$\begin{gathered} \exists m \in E, n = m - k + 1 \\ \Downarrow \\ m = n + k - 1 = \min E \end{gathered}$$

(B) Suppose $E^{\uparrow} \neq \varnothing$. Then

$$\exists k \in \mathbb{Z}, \forall \alpha \in E, \alpha \leq k$$

Define

$$F := \{-\alpha + k + 1 \mid \alpha \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{gathered} F \neq \varnothing \\ \Downarrow \\ \exists n \in F, n = \min F \\ \Downarrow \\ \forall \alpha \in E, n \leq -\alpha + k + 1 \\ \Downarrow \\ \forall \alpha \in E, \alpha \leq k - n + 1 \end{gathered}$$

Theorem 1.1.4

Therefore,

$$\begin{gathered} \exists M \in E, n = -M + k + 1 \\ \Downarrow \\ M = k - n + 1 = \max E \end{gathered}$$

□

Definition 1.4.5. Suppose $E \subseteq |(S, <)|$. Then

- Suppose $E^{\uparrow} \neq \varnothing$. An element $\alpha \in S$ is called a **supremum** of E if

  $$\alpha = \min E^{\uparrow}$$

  Such an α is denoted by

  $$\alpha = \sup E$$

- Suppose $E^{\downarrow} \neq \varnothing$. An element $\alpha \in S$ is called an **infimum** of E if

  $$\alpha = \max E^{\downarrow}$$

  Such an α is denoted by

  $$\alpha = \inf E$$

Theorem 1.4.6. *Define*

$$S := \left\{p \in \mathbb{Q}^+ \;\middle|\; p^2 < 2\right\} \subseteq |(\mathbb{Q}, <)|$$

Then

$$\nexists \alpha \in \mathbb{Q}^+, \alpha = \sup S$$

Proof.

- Suppose $p \in S$. Define

$$q \coloneqq p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2} > p$$

Then

$$\begin{gathered} q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2} < 0 \\ \Downarrow \\ q \in S \\ \Downarrow \\ p \notin S^{\uparrow} \end{gathered}$$

- Suppose $p \in \mathbb{Q}^+ \setminus S$. Then

$$\begin{gathered} p^2 \geq 2 \\ \Downarrow \\ p^2 > 2 \\ \Downarrow \\ p \in S^{\uparrow} \end{gathered} \qquad \nexists p \in \mathbb{Q}^+, p^2 = 2$$

Define

$$q \coloneqq p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2} < p$$

Then

$$\begin{gathered} q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2} > 0 \\ \Downarrow \\ q^2 > 2 \\ \Downarrow \\ q \in S^{\uparrow} \\ \Downarrow \\ p \neq \min S^{\uparrow} \end{gathered}$$

$\square$

Definition 1.4.7. Suppose

$$\forall E \subseteq |(S,<)|, E \neq \varnothing \wedge E^{\uparrow} \neq \varnothing \implies \exists \alpha \in S, \alpha = \sup E$$

Then $(S,<)$ is said to have the **least upper bound property**.

Theorem 1.4.8. *Suppose* $(S, <)$ *have the least upper bound property. Then*

$$\forall E \subseteq |(S, <)|, E \neq \varnothing \wedge E^{\downarrow} \neq \varnothing \implies \exists \alpha \in S, \alpha = \inf E$$

Proof. Suppose

- $E \subseteq |(S, <)|$
- $E \neq \varnothing$
- $E^{\downarrow} \neq \varnothing$

Then

$$\begin{gathered} \exists x \in E \\ \Downarrow \\ \forall y \in E^{\downarrow}, y \leq x \\ \Downarrow \\ \left(E^{\downarrow}\right)^{\uparrow} \neq \varnothing \\ \Downarrow \\ \exists \alpha \in S, \alpha = \sup E^{\downarrow} \end{gathered}$$

Assume $\alpha \notin E^{\downarrow}$. Then

$$\begin{gathered} \exists \beta \in E, \beta < \alpha \\ \Downarrow \qquad \alpha = \sup E^{\downarrow} \\ \beta \notin \left(E^{\downarrow}\right)^{\uparrow} \\ \Downarrow \\ \exists \gamma \in E^{\downarrow}, \beta < \gamma \leq \alpha \\ \Downarrow \\ \forall x \in E, \gamma \leq x \\ \Downarrow \\ \gamma \leq \beta \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} \alpha \in E^{\downarrow} \\ \Downarrow \qquad \alpha = \sup E^{\downarrow} \\ \alpha = \max E^{\downarrow} \\ \Downarrow \\ \alpha = \inf E \end{gathered}$$

$\square$

1.5 Fields and Ordered Fields

Definition 1.5.1. Suppose two operations

- **addition** $+$
- **multiplication** $\cdot$

on F satisfy the following axioms:

(A) Axioms for **addition**

- $\forall x, y \in F, x + y \in F$
- $\forall x, y \in F, x + y = y + x$
- $\forall x, y, z \in F, (x + y) + z = x + (y + z)$
- $\exists 0 \in F, \forall x \in F, x + 0 = x$
- $\forall x \in F, \exists - x \in F, x + (-x) = 0$

(B) Axioms for **multiplication**

- $\forall x, y \in F, x \cdot y \in F$
- $\forall x, y \in F, x \cdot y = y \cdot x$
- $\forall x, y, z \in F, (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- $\exists 1 \in F, 1 \neq 0 \wedge (\forall x \in F, x \cdot 1 = x)$
- $\forall x \in F, x \neq 0 \implies \exists x^{-1} \in F, x \cdot x^{-1} = 1$

(C) The **distributive law**

- $\forall x, y, z \in F, x \cdot (y + z) = x \cdot y + x \cdot z$

Then the structure $(F, +, \cdot)$ is called a **field**.

Definition 1.5.2. Suppose a relation $<$ on $|(F, +, \cdot)|$ satisfies the following axioms:

- $\forall x, y, z \in F, x < y \implies x + z < y + z$
- $\forall x, y \in F, x > 0 \wedge y > 0 \implies x \cdot y > 0$

Then the structure $(F, +, \cdot, <)$ is called an **ordered field**.

Remark. $(\mathbb{Q}, +, \cdot, <)$ is an ordered field. Also, there exists an ordered field $(\mathbb{R}, +, \cdot, <)$ which has the *least upper bound property*. In this textbook, the construction of $\mathbb{R}$ is not discussed, so we will just assume that $(\mathbb{R}, +, \cdot, <)$ exists and satisfies axiom 1.6.1.

1.6 The Real Number System

Axiom 1.6.1. $(\mathbb{R}, +, \cdot, <)$ has the *least upper bound property*.

Theorem 1.6.2. *$(\mathbb{R}, +, \cdot, <)$ has the **Archimedean property**, which states that*

$$\forall x \in \mathbb{R}^+, \forall y \in \mathbb{R}, \exists n \in \mathbb{N}, nx > y$$

where

$$\mathbb{R}^+ := \{x \in \mathbb{R} \mid x > 0\}$$

Proof. Define

$$E := \{nx \mid nx \le y,\ n \in \mathbb{N}\} \subseteq \mathbb{R}$$

Then

- If $E = \varnothing$, then
$$\begin{gathered} 1 \cdot x \notin E \\ \Downarrow \\ 1 \cdot x > y \end{gathered}$$
- Otherwise,
$$\begin{gathered} E \neq \varnothing \wedge y \in E^{\uparrow} \\ \Downarrow \qquad \text{Axiom 1.6.1} \\ \exists \alpha \in \mathbb{R}, \alpha = \sup E \\ \Downarrow \\ \alpha - x \notin E^{\uparrow} \\ \Downarrow \\ \exists m \in \mathbb{N}, mx > \alpha - x \\ \Downarrow \\ (m+1)x > \alpha \\ \Downarrow \qquad \alpha = \sup E \\ (m+1)x \notin E \\ \Downarrow \\ (m+1)x > y \end{gathered}$$

□

Lemma 1.6.3.

$$\forall x \in [-1, +\infty), \forall n \in \mathbb{N}, (1+x)^n \ge 1 + nx$$

*This inequality is called **Bernoulli's inequality**.*

Proof. Suppose $x \in [-1, +\infty)$. Define

$$\forall n \in \mathbb{N}, P(n) : (1+x)^n \geq 1 + nx$$

Then

- **Base Case:**
$$\begin{gathered}(1+x)^1 = 1 + 1 \cdot x \\ \Downarrow \\ P(1) \text{ is true}\end{gathered}$$

- **Inductive step:** Suppose $P(n)$ is true. Then
$$\begin{aligned}
&(1+x)^n \geq 1 + nx \\
&\Downarrow && 1 + x \geq 0 \\
&(1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \\
&\Downarrow && x^2 \geq 0 \\
&(1+x)^{n+1} \geq 1 + (n+1)x \\
&\Downarrow \\
&P(n+1) \text{ is true}
\end{aligned}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n) \text{ is true}$$

□

Theorem 1.6.4.

$$\forall \varepsilon \in \mathbb{R}^+, \forall b \in (1, +\infty), \exists n \in \mathbb{N}, b^{-n} < \varepsilon$$

Proof. Suppose

- $\varepsilon \in \mathbb{R}^+$
- $b \in (1, +\infty)$

Then

$$\begin{aligned}
&b - 1 \in \mathbb{R}^+ \\
&\Downarrow && \text{Theorem 1.6.2} \\
&\exists n \in \mathbb{N}, n(b-1) > \frac{1}{\varepsilon} \\
&\Downarrow && \text{Lemma 1.6.3} \\
&(1 + (b-1))^n \geq 1 + n(b-1) > \frac{1}{\varepsilon} \\
&\Downarrow \\
&(1 + (b-1))^{-n} < \varepsilon \\
&\Downarrow \\
&b^{-n} < \varepsilon
\end{aligned}$$

□

Theorem 1.6.5. *Suppose*

- $E \subseteq \mathbb{Z}$
- $E \neq \varnothing$

Then

(A) $E_{\mathbb{R}}^{\downarrow} \neq \varnothing \implies \exists m \in E, m = \min E$

(B) $E_{\mathbb{R}}^{\uparrow} \neq \varnothing \implies \exists M \in E, M = \max E$

where

- $E_{\mathbb{R}}^{\downarrow} \coloneqq \{\alpha \in \mathbb{R} \mid \forall x \in E, \alpha \leq x\}$
- $E_{\mathbb{R}}^{\uparrow} \coloneqq \{\alpha \in \mathbb{R} \mid \forall x \in E, x \leq \alpha\}$

Proof.

(A) Suppose $E_{\mathbb{R}}^{\downarrow} \neq \varnothing$. Then

$$\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x$$

- If $\alpha \geq 0$, then

$$\begin{gathered} \forall x \in E, 0 \leq \alpha \leq x \\ \Downarrow \\ 0 \in E^{\downarrow} \end{gathered}$$

- Otherwise, by theorem 1.6.2,

$$\begin{gathered} \exists n \in \mathbb{N}, n > -\alpha \\ \Downarrow \\ \forall x \in E, -n < \alpha \leq x \\ \Downarrow \\ -n \in E^{\downarrow} \end{gathered}$$

Therefore,

$$\begin{gathered} E^{\downarrow} \neq \varnothing \\ \Downarrow \qquad \text{Theorem 1.4.4} \\ \exists m \in E, m = \min E \end{gathered}$$

(B) Suppose $E_{\mathbb{R}}^{\uparrow} \neq \varnothing$. Then

$$\begin{gathered} \exists \alpha \in \mathbb{R}, \forall x \in E, x \leq \alpha \\ \Downarrow \qquad \text{Theorem 1.6.2} \\ \exists n \in \mathbb{N}, n > \alpha \\ \Downarrow \\ \forall x \in E, x \leq \alpha < n \\ \Downarrow \\ n \in E^{\uparrow} \\ \Downarrow \qquad \text{Theorem 1.4.4} \\ \exists M \in E, M = \max E \end{gathered}$$

$\square$

Definition 1.6.6. Suppose $x \in \mathbb{R}$.

- The **floor function** $\lfloor x \rfloor$ is defined by

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$$

- The **ceiling function** $\lceil x \rceil$ is defined by

$$\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

Theorem 1.6.5 guarantees the *existence* of $\lfloor x \rfloor$ and $\lceil x \rceil$.

Theorem 1.6.7. *$\mathbb{Q}$ is **dense** in $\mathbb{R}$, which states that*

$$\forall x, y \in \mathbb{R}, x < y \implies \exists p \in \mathbb{Q}, x < p < y$$

Proof. Suppose

- $x, y \in \mathbb{R}$
- $x < y$

Define

$$\varepsilon := y - x > 0$$

By theorem 1.6.2,

$$\exists n \in \mathbb{N}, \frac{1}{n} < \varepsilon$$

Define

$$S := \left\{ k \in \mathbb{Z} \;\middle|\; \frac{k}{n} \leq x \right\} \subseteq \mathbb{Z}$$

Then

$$\begin{gathered} \lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \\ \Downarrow \\ n\lfloor x \rfloor \leq nx < n\lfloor x \rfloor + n \\ \Downarrow \\ n\lfloor x \rfloor \in S \end{gathered}$$

Also,

$$nx \in S_{\mathbb{R}}^{\uparrow}$$

By theorem 1.6.5,

$$\exists M \in \mathbb{Z}, M = \max S$$

Then

$$\begin{gathered} M \in S \wedge M+1 \notin S \\ \Downarrow \\ \frac{M}{n} \leq x < \frac{M+1}{n} \\ \Downarrow \\ x < \frac{M+1}{n} \leq x + \frac{1}{n} < x + \varepsilon = x + (y - x) = y \end{gathered}$$

□

Lemma 1.6.8.

$$\forall n \in \mathbb{N}, \forall a, b \in \mathbb{R}^+, a < b \implies b^n - a^n \leq (b-a) n b^{n-1}$$

Proof. Suppose

- $n \in \mathbb{N}$
- $a, b \in \mathbb{R}^+$
- $a < b$

Then

$$b^n - a^n = (b-a) \sum_{k=0}^{n-1} b^{n-1-k} a^k \leq (b-a) n b^{n-1}$$

□

Lemma 1.6.9.

$$\forall n \in \mathbb{N}, \forall a, b \in \mathbb{R}^+, a < b \iff a^n < b^n$$

Proof. Suppose

- $n \in \mathbb{N}$
- $a, b \in \mathbb{R}^+$

Then

$$\begin{gathered} b^n - a^n = (b-a) \sum_{k=0}^{n-1} b^{n-1-k} a^k \\ \Downarrow \\ a < b \iff a^n < b^n \end{gathered}$$

□

Theorem 1.6.10.

$$\forall x \in \mathbb{R}^+, \forall n \in \mathbb{N}, \exists! y \in \mathbb{R}^+, y^n = x$$

Such an y is denoted by

- $y = \sqrt[n]{x}$
- $y = x^{\frac{1}{n}}$

Proof. Suppose

- $x \in \mathbb{R}^+$
- $n \in \mathbb{N}$

Define

$$E := \left\{ t \in \mathbb{R}^+ \;\middle|\; t^n < x \right\} \subseteq \mathbb{R}$$

- If $x \geq 1$, then

$$\begin{gathered} 2^{-n} < 1 \leq x \\ \Downarrow \\ 2^{-1} \in E \end{gathered}$$

  Also,

$$\begin{gathered} \forall t \in E, t^n < x \leq x^n \\ \Downarrow \\ \forall t \in E, t < x \\ \Downarrow \\ x \in E^{\uparrow} \end{gathered} \qquad \text{Lemma 1.6.9}$$

- Otherwise,

$$\begin{gathered} x < 1 \\ \Downarrow \\ x^{2n} < x \\ \Downarrow \\ x^2 \in E \end{gathered}$$

  Also,

$$\begin{gathered} \forall t \in E, t^n < x < 1 = 1^n \\ \Downarrow \\ \forall t \in E, t < 1 \\ \Downarrow \\ 1 \in E^{\uparrow} \end{gathered} \qquad \text{Lemma 1.6.9}$$

Therefore,

$$\begin{gathered} E \neq \varnothing \wedge E^{\uparrow} \neq \varnothing \\ \Downarrow \\ \exists \alpha \in \mathbb{R}^+, \alpha = \sup E \end{gathered} \qquad \text{Axiom 1.6.1}$$

We will show

$$\begin{gathered} \alpha^n \not< x \wedge \alpha^n \not> x \\ \Downarrow \\ \alpha^n = x \end{gathered}$$

- Assume $\alpha^n < x$. By theorem 1.6.7,

$$\begin{gathered} \exists h \in \mathbb{Q}, 0 < h < \min\left\{1, \frac{x-\alpha^n}{n(\alpha+1)^{n-1}}\right\} \\ \Downarrow \qquad \text{Lemma 1.6.8} \\ (\alpha+h)^n - \alpha^n \leq hn(\alpha+h)^{n-1} < hn(\alpha+1)^{n-1} \\ \Downarrow \\ (\alpha+h)^n < \alpha^n + hn(\alpha+1)^{n-1} < \alpha^n + (x-\alpha^n) = x \\ \Downarrow \\ \alpha + h \in E \\ \Downarrow \qquad \alpha = \sup E \\ \perp \end{gathered}$$

- Assume $\alpha^n > x$. By theorem 1.6.7,

$$\begin{gathered} \exists h \in \mathbb{Q}, 0 < h < \min\left\{\alpha, \frac{\alpha^n - x}{n\alpha^{n-1}}\right\} \\ \Downarrow \qquad \text{Lemma 1.6.8} \\ \alpha^n - (\alpha-h)^n \leq hn\alpha^{n-1} \\ \Downarrow \\ (\alpha-h)^n \geq \alpha^n - hn\alpha^{n-1} > \alpha^n - (\alpha^n - x) = x \\ \Downarrow \\ \forall t \in E, t^n < x < (\alpha-h)^n \\ \Downarrow \qquad \text{Lemma 1.6.9} \\ \forall t \in E, t < \alpha - h \\ \Downarrow \\ \alpha - h \in E^{\uparrow} \\ \Downarrow \qquad \alpha = \sup E \\ \perp \end{gathered}$$

This guarantees the *existence* of y. Let

- $y_1, y_2 \in \mathbb{R}^+$
- $(y_1)^n = x$
- $(y_2)^n = x$

Assume $y_1 \neq y_2$ (*WLOG* $y_1 < y_2$). Then

$$\begin{array}{c} y_1 < y_2 \\ \Downarrow \\ (y_1)^n < (y_2)^n \\ \Downarrow \\ x < x \\ \perp \end{array} \qquad \text{Lemma 1.6.9}$$

Therefore,

$$y_1 = y_2$$

This guarantees the *uniqueness* of y. $\square$

Definition 1.6.11. Suppose

- $y \in [0, 1)$
- $b \in [2, \infty)_{\mathbb{N}}$

A sequence $(a_n)_{n=1}^{\infty}$ in $[0, b-1]_{\mathbb{N}}$ is called a **greedy base b expansion** of y if

$$\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$$

Such an $(a_n)_{n=1}^{\infty}$ is denoted by

$$(a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$$

Theorem 1.6.12. *Suppose $b \in [2, \infty)_{\mathbb{N}}$. Then*

$$\forall y \in [0, 1), \exists!(a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}, (a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$$

Proof. Suppose $y \in [0, 1)$. Define

- $(t_n)_{n=1}^{\infty}$ in $[0, b)$
- $(T_n)_{n=1}^{\infty}$ in $\mathcal{P}([0, b-1]_{\mathbb{N}})$
- $(a_n)_{n=1}^{\infty}$ in $[0, b-1]_{\mathbb{N}}$

as

$$
\begin{array}{ccc}
t_1 := by & T_1 := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_1\} & a_1 := \max T_1 \\
t_2 := b^2 y - ba_1 & T_2 := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_2\} & a_2 := \max T_2 \\
\vdots & \vdots & \vdots \\
t_n := b^n y - \displaystyle\sum_{k=1}^{n-1} a_k b^{n-k} & T_n := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_n\} & a_n := \max T_n \\
\vdots & \vdots & \vdots
\end{array}
$$

Define

$$\forall n \in \mathbb{N}, P(n) : \forall k \in [1, n]_{\mathbb{N}}, 0 \leq t_k < b$$

Then

- **Base case:**
$$
\begin{gathered}
0 \leq y < 1 \\
\Downarrow \\
0 \leq t_1 = by < b \\
\Downarrow \\
P(1) \text{ is true}
\end{gathered}
$$

- **Inductive step:** Suppose $P(n)$ is true. Then
$$
\begin{gathered}
\forall k \in [1, n]_{\mathbb{N}}, 0 \in T_k \wedge b-1 \in (T_k)^{\uparrow} \\
\Downarrow \qquad \text{Theorem 1.4.4} \\
\forall k \in [1, n]_{\mathbb{N}}, \exists M \in [0, b-1]_{\mathbb{N}}, M = \max T_k
\end{gathered}
$$
Therefore, $a_1, a_2, \ldots, a_n$ are well-defined. Then
$$
\begin{aligned}
t_{n+1} &= b^{n+1} y - \sum_{k=1}^{n} a_k b^{n+1-k} \\
&= b\left(b^n y - \sum_{k=1}^{n} a_k b^{n-k}\right) \\
&= b\left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n\right) \\
&= b(t_n - a_n)
\end{aligned}
$$
Since $P(n)$ is true,
$$
\begin{gathered}
a_n \leq t_n < a_n + 1 \\
\Downarrow \\
0 \leq t_n - a_n < 1 \\
\Downarrow \\
0 \leq t_{n+1} = b(t_n - a_n) < b \\
\Downarrow \\
P(n+1) \text{ is true}
\end{gathered}
$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n) \text{ is true}$$

Therefore, $(t_n)_{n=1}^{\infty}$ is well-defined. Then

$$\begin{gathered} \forall n \in \mathbb{N}, 0 \in T_n \wedge b-1 \in (T_n)^{\uparrow} \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists M \in [0, b-1]_{\mathbb{N}}, M = \max T_n \end{gathered} \qquad \text{Theorem 1.4.4}$$

Therefore, $(a_n)_{n=1}^{\infty}$ is well-defined. Then

$$\begin{gathered} \forall n \in \mathbb{N}, a_n \leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \Downarrow \\ \forall n \in \mathbb{N}, 0 \leq b^n y - \sum_{k=1}^{n} a_k b^{n-k} < 1 \\ \Downarrow \\ \forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \end{gathered}$$

This guarantees the *existence* of $\text{Greedy}_b(y)$. Let

- $(a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$
- $(c_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$

Define

$$\forall n \in \mathbb{N}, Q(n) : \forall k \in [1, n]_{\mathbb{N}}, a_k = c_k$$

Then

- **Base case:** By definition 1.6.11,

$$\begin{gathered} 0 \leq y - \frac{a_1}{b} < \frac{1}{b} \\ \Downarrow \\ \frac{a_1}{b} \leq y < \frac{a_1+1}{b} \end{gathered} \qquad \begin{gathered} 0 \leq y - \frac{c_1}{b} < \frac{1}{b} \\ \Downarrow \\ \frac{c_1}{b} \leq y < \frac{c_1+1}{b} \end{gathered}$$

Then

$$\begin{gathered} \frac{a_1}{b} \leq y < \frac{c_1+1}{b} \\ \Downarrow \\ a_1 < c_1 + 1 \\ \Downarrow \\ a_1 \leq c_1 \end{gathered} \qquad \begin{gathered} \frac{c_1}{b} \leq y < \frac{a_1+1}{b} \\ \Downarrow \\ c_1 < a_1 + 1 \\ \Downarrow \\ c_1 \leq a_1 \end{gathered}$$

Therefore,

$$\begin{gathered} a_1 = c_1 \\ \Downarrow \\ Q(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Suppose $Q(n)$ is true. By definition 1.6.11,

$$\begin{gathered} 0 \le y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \end{gathered} \qquad \begin{gathered} 0 \le y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \end{gathered}$$

Then

$$\begin{gathered} \sum_{k=1}^{n+1} \frac{a_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}} \end{gathered} \qquad \begin{gathered} \sum_{k=1}^{n+1} \frac{c_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}} \end{gathered}$$

Since $Q(n)$ is true,

$$\begin{gathered} \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} \\ \Downarrow \\ a_{n+1} - c_{n+1} < 1 \\ \Downarrow \\ a_{n+1} \le c_{n+1} \end{gathered} \qquad \begin{gathered} \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} \\ \Downarrow \\ c_{n+1} - a_{n+1} < 1 \\ \Downarrow \\ c_{n+1} \le a_{n+1} \end{gathered}$$

Therefore,

$$\begin{gathered} a_{n+1} = c_{n+1} \\ \Downarrow \\ Q(n+1) \text{ is true} \end{gathered}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, Q(n) \text{ is true}$$

This guarantees the *uniqueness* of $\text{Greedy}_b(y)$. □

Theorem 1.6.13. *Suppose*

- $b \in [2, \infty)_{\mathbb{N}}$

- $a, c \in [0, 1)$
- $(a_n)_{n=1}^{\infty} = \text{Greedy}_b(a)$
- $(c_n)_{n=1}^{\infty} = \text{Greedy}_b(c)$

Then

$$(\forall n \in \mathbb{N}, a_n = c_n) \implies a = c$$

Proof. Suppose

$$\forall n \in \mathbb{N}, a_n = c_n$$

Assume $a \neq c$ (*WLOG* $a < c$). By theorem 1.6.4,

$$\exists n \in \mathbb{N}, \frac{1}{b^n} < c - a$$

By definition 1.6.11,

$$\begin{array}{cc} 0 \leq a - \displaystyle\sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} & 0 \leq c - \displaystyle\sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n} \\ \Downarrow & \Downarrow \\ \displaystyle\sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} & c < \displaystyle\sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n} \end{array}$$

Then

$$\begin{gathered} \sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n} \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$a = c$$

$\square$

Theorem 1.6.14. *Suppose*

- $b \in [2, \infty)_{\mathbb{N}}$
- $(a_n)_{n=1}^{\infty}$ *in* $[0, b-1]_{\mathbb{N}}$

Then

$$(\nexists y \in [0, 1), (a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)) \implies \exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = b - 1$$

Proof. We will prove the contrapositive. Suppose

$$\forall N \in \mathbb{N}, \exists n \in [N,\infty)_{\mathbb{N}}, a_n \neq b-1$$

Define

$$\forall n \in \mathbb{N}, x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

Then

- Suppose $n, m \in \mathbb{N}$. Then

$$\begin{gathered} \forall k \in \mathbb{N}, a_k \geq 0 \\ \Downarrow \\ n < m \implies x_n \leq x_m \end{gathered}$$

- Suppose
  - $n, m \in \mathbb{N}$
  - $n > m$

  Then

$$\begin{gathered} x_n = x_m + \sum_{k=m+1}^{n} \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^{n} \frac{b-1}{b^k} \\ \Downarrow \\ x_n \leq x_m + \frac{1}{b^m} - \frac{1}{b^n} \\ \Downarrow \\ x_n + \frac{1}{b^n} \leq x_m + \frac{1}{b^m} \end{gathered}$$

Let $N = 1$. Then

$$\begin{gathered} \exists m \in \mathbb{N}, a_m \neq b-1 \\ \Downarrow \\ a_m + 1 \leq b-1 \end{gathered}$$

This implies

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m+1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^{m} \frac{b-1}{b^k} = 1 - \frac{1}{b^m} \end{aligned}$$

Then

- $\forall n \in [1, m]_{\mathbb{N}}, x_n \leq x_m$
- $\forall n \in [m+1, \infty)_{\mathbb{N}}, x_n < x_m + \dfrac{1}{b^m} \leq 1 - \dfrac{1}{b^m}$

Therefore,

$$\begin{array}{c} 1 - \dfrac{1}{b^m} \in (\{x_n\}_{n=1}^{\infty})^{\uparrow} \\ \Downarrow \\ \exists y \in \mathbb{R}, y = \sup\{x_n\}_{n=1}^{\infty} \\ \Downarrow \\ y \leq 1 - \dfrac{1}{b^m} < 1 \\ \Downarrow \\ y \in [0, 1) \end{array} \qquad \begin{array}{r} \text{Axiom 1.6.1} \\ \\ 0 \leq x_1 \leq y \end{array}$$

Suppose $k \in \mathbb{N}$. Then

$$\begin{array}{c} \exists m \in [k+1, \infty)_{\mathbb{N}}, a_m \neq b - 1 \\ \Downarrow \\ x_m = x_k + \displaystyle\sum_{n=k+1}^{m} \frac{a_n}{b^n} < x_k + \sum_{n=k+1}^{m} \frac{b-1}{b^n} \\ \Downarrow \\ x_m < x_k + \dfrac{1}{b^k} - \dfrac{1}{b^m} \\ \Downarrow \\ x_m + \dfrac{1}{b^m} < x_k + \dfrac{1}{b^k} \end{array}$$

This implies

$$\begin{array}{c} x_m + \dfrac{1}{b^m} \in (\{x_n\}_{n=1}^{\infty})^{\uparrow} \\ \Downarrow \\ y \leq x_m + \dfrac{1}{b^m} < x_k + \dfrac{1}{b^k} \\ \Downarrow \\ 0 \leq y - x_k < \dfrac{1}{b^k} \end{array} \qquad y = \sup\{x_n\}_{n=1}^{\infty}$$

Since k was arbitrary,

$$(a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$$

□

Theorem 1.6.15. *Suppose*

- $b \in [2, \infty)_{\mathbb{N}}$

- $(a_n)_{n=1}^{\infty}$ *in* $[0, b-1]$

Then

$$(\exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = b-1) \implies \nexists y \in [0,1), (a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$$

This theorem is the ***converse*** *of theorem 1.6.14.*

Proof. Suppose

$$\exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = b-1$$

Assume

$$\exists y \in [0,1), (a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$$

Define

$$\forall n \in \mathbb{N}, x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

Then

$$\forall n \in [N+1, \infty)_{\mathbb{N}}, x_n = x_N + \sum_{k=N+1}^{n} \frac{b-1}{b^k} = x_N + \frac{1}{b^N} - \frac{1}{b^n}$$
$$\Downarrow$$
$$\forall n \in [N+1, \infty)_{\mathbb{N}}, y - x_n = y - x_N - \frac{1}{b^N} + \frac{1}{b^n} \geq 0$$
$$\Downarrow$$
$$\forall n \in [N+1, \infty)_{\mathbb{N}}, \frac{1}{b^n} \geq x_N + \frac{1}{b^N} - y$$

By definition 1.6.11,

$$0 \leq y - x_N < \frac{1}{b^N}$$
$$\Downarrow$$
$$x_N + \frac{1}{b^N} - y > 0$$
$$\Downarrow \qquad \text{Theorem 1.6.4}$$
$$\exists m \in \mathbb{N}, \frac{1}{b^m} < x_N + \frac{1}{b^N} - y$$
$$\Downarrow$$
$$\forall n \in [\max\{m, N\} + 1, \infty)_{\mathbb{N}}, \frac{1}{b^n} < \frac{1}{b^m} < x_N + \frac{1}{b^N} - y$$
$$\Downarrow$$
$$\perp$$

Therefore,

$$\nexists y \in [0,1), (a_n)_{n=1}^{\infty} = \text{Greedy}_b(y)$$

□

1.7 The Extended Real Number System

Definition 1.7.1. The extended real number system $\overline{\mathbb{R}}$ is defined by

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$$

$(\overline{\mathbb{R}}, <)$ is ordered with

$$\forall x \in \mathbb{R}, -\infty < x < +\infty$$

Definition 1.7.2. Operations on $\overline{\mathbb{R}}$ are defined as

- $\forall x \in (-\infty, +\infty], x + (+\infty) = (+\infty) + x := +\infty$
- $\forall x \in [-\infty, +\infty), x + (-\infty) = (-\infty) + x := -\infty$
- $\forall x \in (0, +\infty], x \cdot (\pm\infty) = (\pm\infty) \cdot x := \pm\infty$
- $\forall x \in [-\infty, 0), x \cdot (\pm\infty) = (\pm\infty) \cdot x := \mp\infty$
- $\forall x \in \mathbb{R}, \dfrac{x}{\pm\infty} := 0$

Undefined operations in $\overline{\mathbb{R}}$ are

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\dfrac{(\pm\infty)}{(\pm\infty)}, \dfrac{(\pm\infty)}{(\mp\infty)}$
- $\forall x \in \overline{\mathbb{R}}, \dfrac{x}{0}$

Theorem 1.7.3. *Suppose $E \subseteq \overline{\mathbb{R}}$. Then*

(A) $\exists x \in \overline{\mathbb{R}}, x = \sup E$

(B) $\exists y \in \overline{\mathbb{R}}, y = \inf E$

Proof. Suppose $E \subseteq \overline{\mathbb{R}}$.

- If $E = \varnothing$, then

  (A) $\sup E = -\infty$
  (B) $\inf E = +\infty$

- If $E = \{-\infty\}$, then

(A) $\sup E = -\infty$

(B) $\inf E = -\infty$

- If $E = \{+\infty\}$, then

  (A) $\sup E = +\infty$

  (B) $\inf E = +\infty$

- Otherwise,

  (A) – If $E^{\uparrow}_{\mathbb{R}} \neq \varnothing$, then

$$\begin{array}{c} E \setminus \{-\infty\} \subseteq \mathbb{R} \\ \Downarrow \\ \exists x \in \mathbb{R}, x = \sup E \setminus \{-\infty\} \\ \Downarrow \\ x = \sup E \end{array}$$

Axiom 1.6.1

  – Otherwise,

$$\sup E = +\infty$$

  (B) – If $E^{\downarrow}_{\mathbb{R}} \neq \varnothing$, then

$$\begin{array}{c} E \setminus \{+\infty\} \subseteq \mathbb{R} \\ \Downarrow \\ \exists x \in \mathbb{R}, x = \inf E \setminus \{+\infty\} \\ \Downarrow \\ x = \inf E \end{array}$$

Theorem 1.4.8

  – Otherwise,

$$\inf E = -\infty$$

□

1.8 The Complex Number System

Definition 1.8.1. The complex number system $\mathbb{C}$ is defined by

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}$$

Definition 1.8.2. The operations on $\mathbb{C}$ are defined as

- $\forall (x, y), (u, v) \in \mathbb{C}, (x, y) + (u, v) := (x + u, y + v)$
- $\forall (x, y), (u, v) \in \mathbb{C}, (x, y) \cdot (u, v) := (xu - yv, xv + yu)$

Remark. One can verify that $(\mathbb{C}, +, \cdot)$ is a field with

- The **additive identity** is $(0, 0)$
- The **multiplicative identity** is $(1, 0)$
- The **additive inverse** of (x, y) is $(-x, -y)$
- The **multiplicative inverse** of $(x, y) \neq (0, 0)$ is

$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right)$$

Definition 1.8.3. The **imaginary unit** i is defined by

$$i := (0, 1) \in \mathbb{C}$$

Then

$$i^2 = (0, 1) \cdot (0, 1) = (-1, 0)$$

Remark. There is a **canonical embedding**

$$\mathbb{R} \to \mathbb{C} \text{ such that } x \mapsto (x, 0)$$

Therefore, we identify $\mathbb{R}$ with its image in $\mathbb{C}$ and write

$$\mathbb{R} \subseteq \mathbb{C}$$

Then

$$\begin{aligned} \forall (x, y) \in \mathbb{C}, (x, y) &= (x, 0) + (0, y) \\ &= (x, 0) + (y, 0) \cdot (0, 1) \\ &= x + yi \end{aligned}$$

Definition 1.8.4. Suppose $z = (x, y) \in \mathbb{C}$.

- The **real part** of z is defined by
  - $\Re(z) := x$
  - $\mathrm{Re}(z) := x$
- The **imaginary part** of z is defined by
  - $\Im(z) := y$

  - $\mathrm{Im}(z) := y$

- The **complex conjugate** of z is defined by

$$\overline{z} := (x, -y) = x - yi$$

Theorem 1.8.5. *Suppose*

- $z = (x, y) \in \mathbb{C}$
- $w = (u, v) \in \mathbb{C}$

Then

- $\overline{(\overline{z})} = z$
- $z = \overline{z} \iff z \in \mathbb{R}$
- $\overline{z + w} = \overline{z} + \overline{w}$
- $\overline{zw} = \overline{z} \cdot \overline{w}$
- $z\overline{z} = x^2 + y^2 \in [0, \infty)$

Proof. The proof follows directly from definitions 1.8.2 and 1.8.4. □

Definition 1.8.6. Suppose $z = (x, y) \in \mathbb{C}$. The **modulus** of z is defined by

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in [0, \infty)$$

Theorem 1.6.10 guarantees the *existence* of $|z|$.

Theorem 1.8.7. *Suppose*

- $n \in \mathbb{N}$
- $a_1, a_2, \ldots, a_n \in \mathbb{C}$
- $b_1, b_2, \ldots, b_n \in \mathbb{C}$

Then

$$\left| \sum_{k=1}^{n} a_k \overline{b_k} \right|^2 \leq \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2$$

This theorem is called the ***Cauchy-Schwarz inequality****.*

Proof. Suppose $i, j \in [1, n]_{\mathbb{N}}$. Then

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i)\overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) \end{aligned}$$

Consider

$$\begin{aligned} \sum_{i<j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i<j} (|a_i|^2 |b_j|^2) + \sum_{j<i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2 \end{aligned}$$

Also,

$$\begin{aligned} \sum_{i<j} (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) &= \sum_{i \neq j} (a_i \overline{b_i} \cdot \overline{a_j} b_j) \\ &= \sum_{i=1}^{n} a_i \overline{b_i} \cdot \sum_{j=1}^{n} \overline{a_j} b_j - \sum_{k=1}^{n} a_k \overline{b_k} \cdot \overline{a_k} b_k \\ &= \left| \sum_{k=1}^{n} a_k \overline{b_k} \right|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2 \end{aligned}$$

Therefore,

$$\sum_{i<j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \left| \sum_{k=1}^{n} a_k \overline{b_k} \right|^2 \geq 0$$

□

Corollary 1.8.8. *Suppose*

- $n \in \mathbb{N}$
- $a_1, a_2, \ldots, a_n \in \mathbb{R}$
- $b_1, b_2, \ldots, b_n \in \mathbb{R}$

Then

$$\left(\sum_{k=1}^{n} a_k b_k \right)^2 \leq \sum_{k=1}^{n} a_k^2 \cdot \sum_{k=1}^{n} b_k^2$$

Theorem 1.8.9. *Suppose* $z, w \in \mathbb{C}$. *Then*

(A) $|zw| = |z| \cdot |w|$

(B) $|z + w| \leq |z| + |w|$

(C) $||z| - |w|| \leq |z - w|$

Proof. Suppose

- $z = (x, y) \in \mathbb{C}$
- $w = (u, v) \in \mathbb{C}$

By corollary 1.8.8,

$$(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$$

Then

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$

(B)

$$\begin{aligned} |z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2 \end{aligned}$$

(C)

$$\begin{aligned} |z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2 \end{aligned}$$

□

1.9 Euclidean Spaces

Definition 1.9.1. Suppose $n \in \mathbb{N}$. The **Euclidean** n**-space** $\mathbb{R}^n$ is defined by

$$\mathbb{R}^n := \{\boldsymbol{x} = (x_1, x_2, \ldots, x_n) \mid \forall k \in [1, n]_{\mathbb{N}}, x_k \in \mathbb{R}\}$$

$x_1, x_2, \ldots, x_n$ are called the **coordinates** of $\boldsymbol{x}$. The elements of $\mathbb{R}^n$ for $n > 1$ are called **points** and are represented by bold symbols.

Definition 1.9.2. The operations on $\mathbb{R}^n$ are defined as

- $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{x} + \boldsymbol{y} := (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$
- $\forall \alpha \in \mathbb{R}, \forall \boldsymbol{x} \in \mathbb{R}^n, \alpha \boldsymbol{x} := (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)$

Definition 1.9.3. Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.

- The **inner product** of $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined by

$$\boldsymbol{x} \cdot \boldsymbol{y} := \sum_{k=1}^{n} x_k y_k$$

- The **norm** of $\boldsymbol{x}$ is defined by

$$\|\boldsymbol{x}\| := \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{\sum_{k=1}^{n} x_k^2}$$

Theorem 1.9.4. *Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then*

(A) $|\boldsymbol{x} \cdot \boldsymbol{y}| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$

(B) $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$

(C) $\|\boldsymbol{x} - \boldsymbol{y}\| \geq |\|\boldsymbol{x}\| - \|\boldsymbol{y}\||$

Proof.

(A) By corollary 1.8.8,

$$|\boldsymbol{x} \cdot \boldsymbol{y}|^2 = \left|\sum_{k=1}^{n} x_k y_k\right|^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} y_k^2 = \|\boldsymbol{x}\|^2 \cdot \|\boldsymbol{y}\|^2$$

(B) By part **(A)**,

$$\begin{aligned}
\|\boldsymbol{x} + \boldsymbol{y}\|^2 &= (\boldsymbol{x} + \boldsymbol{y}) \cdot (\boldsymbol{x} + \boldsymbol{y}) \\
&= \|\boldsymbol{x}\|^2 + 2\boldsymbol{x} \cdot \boldsymbol{y} + \|\boldsymbol{y}\|^2 \\
&\leq \|\boldsymbol{x}\|^2 + 2|\boldsymbol{x} \cdot \boldsymbol{y}| + \|\boldsymbol{y}\|^2 \\
&\leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 \\
&= (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2
\end{aligned}$$

(C) By part **(B)**,

- If $\|\boldsymbol{x}\| \geq \|\boldsymbol{y}\|$, then

$$\begin{gathered}\|\boldsymbol{x}\| = \|(\boldsymbol{x}-\boldsymbol{y})+\boldsymbol{y}\| \leq \|\boldsymbol{x}-\boldsymbol{y}\| + \|\boldsymbol{y}\| \\ \Downarrow \\ \|\boldsymbol{x}\| - \|\boldsymbol{y}\| \leq \|\boldsymbol{x}-\boldsymbol{y}\|\end{gathered}$$

- Otherwise,

$$\begin{gathered}\|\boldsymbol{y}\| = \|(\boldsymbol{y}-\boldsymbol{x})+\boldsymbol{x}\| \leq \|\boldsymbol{y}-\boldsymbol{x}\| + \|\boldsymbol{x}\| \\ \Downarrow \\ \|\boldsymbol{y}\| - \|\boldsymbol{x}\| \leq \|\boldsymbol{x}-\boldsymbol{y}\|\end{gathered}$$

□

Chapter 2

Basic Topology

2.1 Functions

Definition 2.1.1. Suppose a relation G from X to Y satisfies

$$\forall x \in X, \exists! y \in Y, (x, y) \in G$$

Then the ordered triple

$$f \coloneqq (X, Y, G)$$

is called a **function**, where

- X is called the **domain** of f
- Y is called the **codomain** of f
- G is called the **graph** of f

The notation

- $f : X \to Y$ denotes that f is a function from X to Y
- $f(x) = y$ denotes $(x, y) \in G$

Definition 2.1.2. Suppose $f : X \to Y$.

- The **image** of $E \subseteq X$ under f is defined by

$$f(E) \coloneqq \{f(x) \mid x \in E\} \subseteq Y$$

- The **inverse image** of $E \subseteq Y$ under f is defined by

$$f^{-1}(E) \coloneqq \{x \in X \mid f(x) \in E\} \subseteq X$$

Definition 2.1.3. Suppose $f : X \to Y$.

- f is said to be **injective** if
$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$$
The notation $f : X \hookrightarrow Y$ denotes that f is injective.
- f is said to be **surjective** if
$$\forall y \in Y, \exists x \in X, f(x) = y$$
or equivalently,
$$f(X) = Y$$
The notation $f : X \twoheadrightarrow Y$ denotes that f is surjective.
- f is said to be **bijective** if
  - f is injective
  - f is surjective

  or equivalently,
$$\forall y \in Y, \exists! x \in X, f(x) = y$$
The notation $f : X \xrightarrow{\sim} Y$ denotes that f is bijective.

Definition 2.1.4. Suppose $f : X \xrightarrow{\sim} Y$. The **inverse function** $f^{-1} : Y \to X$ of f is defined by
$$\forall (y, x) \in Y \times X, f^{-1}(y) = x \iff f(x) = y$$
By definition 2.1.3,
$$\begin{gathered} \forall y \in Y, \exists! x \in X, f(x) = y \\ \Downarrow \\ \forall y \in Y, \exists! x \in X, f^{-1}(y) = x \end{gathered}$$
Therefore, f^{-1} is well-defined.

Theorem 2.1.5. *Suppose $f : X \xrightarrow{\sim} Y$. Then*
$$f^{-1} : Y \xrightarrow{\sim} X$$

Proof. By definition 2.1.1,
$$\begin{gathered} \forall x \in X, \exists! y \in Y, f(x) = y \\ \Downarrow \\ \forall x \in X, \exists! y \in Y, f^{-1}(y) = x \end{gathered}$$
□

Definition 2.1.6. Suppose

- $f : X \to Y$
- $E \subseteq X$
- $F \subseteq Y$

Then

- The **restriction** $f|_E : E \to Y$ of f to E is defined by

$$\forall x \in E, f|_E(x) = f(x)$$

- The **corestriction** $f|^F : X \to F$ of f to $F \supseteq f(X)$ is defined by

$$\forall x \in X, f|^F(x) = f(x)$$

- The **bi-restriction** $f|_E^F : E \to F$ of f to E and $F \supseteq f(E)$ is defined by

$$\forall x \in E, f|_E^F(x) = f(x)$$

Theorem 2.1.7. *Suppose*

- $f : X \to Y$
- $E \subseteq X$
- $f(E) \subseteq F \subseteq Y$

Then

(A) $f : X \hookrightarrow Y \implies f|_E^F : E \hookrightarrow F$

(B) $f(E) = F \implies f|_E^F : E \twoheadrightarrow F$

(C) $f : X \hookrightarrow Y \wedge f(E) = F \implies f|_E^F : E \xrightarrow{\sim} F$

Proof.

(A) Suppose

- $f : X \hookrightarrow Y$
- $x_1, x_2 \in E$
- $f|_E^F(x_1) = f|_E^F(x_2)$

By definition 2.1.3,

$$\begin{gathered} f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{gathered}$$

(B) Suppose $f(E) = F$. By definition 2.1.2,

$$\begin{gathered} \forall y \in F, \exists x \in E, f(x) = y \\ \Downarrow \\ \forall y \in F, \exists x \in E, f|_E^F(x) = y \end{gathered}$$

(C) follows from **(A)** and **(B)**.

□

Definition 2.1.8. Suppose

- $f : X \to Y$
- $g : Y \to Z$

The **composition** $g \circ f : X \to Z$ of g with f is defined by

$$\forall x \in X, (g \circ f)(x) = g(f(x))$$

Theorem 2.1.9. *Suppose*

- $f : X \to Y$
- $g : Y \to Z$

Then

(A) $f : X \hookrightarrow Y \wedge g : Y \hookrightarrow Z \implies g \circ f : X \hookrightarrow Z$

(B) $f : X \twoheadrightarrow Y \wedge g : Y \twoheadrightarrow Z \implies g \circ f : X \twoheadrightarrow Z$

(C) $f : X \xrightarrow{\sim} Y \wedge g : Y \xrightarrow{\sim} Z \implies g \circ f : X \xrightarrow{\sim} Z$

Proof.

(A) Suppose

- $f : X \hookrightarrow Y$
- $g : Y \hookrightarrow Z$

- $x_1, x_2 \in X$
- $(g \circ f)(x_1) = (g \circ f)(x_2)$

By definition 2.1.3,

$$\begin{gathered} f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{gathered}$$

(B) Suppose

- $f : X \twoheadrightarrow Y$
- $g : Y \twoheadrightarrow Z$

By definition 2.1.3,

- $\forall z \in Z, \exists y \in Y, g(y) = z$
- $\forall y \in Y, \exists x \in X, f(x) = y$

Therefore,

$$\forall z \in Z, \exists x \in X, (g \circ f)(x) = z$$

(C) follows from **(A)** and **(B)**.

□

2.2 Finite Sets

Theorem 2.2.1. *Define* $\sim$ *by*

$$\forall X \forall Y, X \sim Y \iff \exists f : X \xrightarrow{\sim} Y$$

Then $\sim$ *is an equivalence relation.*

Proof. By definition 1.2.2,

- **Reflexivity**: Define $f : X \to X$ by

  $$\forall x \in X, f(x) = x$$

  Then

  - **Injectivity**: Suppose
    - $*$ $x_1, x_2 \in X$
    - $*$ $f(x_1) = f(x_2)$

    Then

    $$x_1 = x_2$$

  - **Surjectivity**: Suppose $y \in X$. Then

$$f(y) = y$$

  Therefore,

$$\begin{gathered} f : X \xrightarrow{\sim} X \\ \Downarrow \\ X \sim X \end{gathered}$$

- **Symmetry**: Suppose $X \sim Y$. Then

$$\begin{gathered} \exists f : X \xrightarrow{\sim} Y \\ \Downarrow \\ f^{-1} : Y \xrightarrow{\sim} X \\ \Downarrow \\ Y \sim X \end{gathered} \qquad \text{Theorem 2.1.5}$$

- **Transitivity**: Suppose
  - $X \sim Y$
  - $Y \sim Z$

  Then
  - $\exists f : X \xrightarrow{\sim} Y$
  - $\exists g : Y \xrightarrow{\sim} Z$

  By theorem 2.1.9,

$$\begin{gathered} g \circ f : X \xrightarrow{\sim} Z \\ \Downarrow \\ X \sim Z \end{gathered}$$

□

Remark. Since there is no **set of all sets** in the underlying set theory, the relation defined in theorem 2.2.1 is a **proper-class relation** rather than a set. Its restriction to any set S becomes an equivalence relation on S.

Definition 2.2.2. Recall the relation $\sim$ defined in theorem 2.2.1. A set X is said to be **finite** if

$$\exists n \in [0, \infty)_{\mathbb{N}}, [1, n]_{\mathbb{N}} \sim X$$

Such an n is called a **cardinality** of X, and is denoted by

$$|X| = n$$

The notation $|X| < \aleph_0$ denotes that X is finite.

Definition 2.2.3. A set X is said to be

- **infinite** if $\neg(|X| < \aleph_0)$
- **countable** if $\mathbb{N} \sim X$
- **at most countable** if $|X| < \aleph_0 \vee \mathbb{N} \sim X$
- **uncountable** if $\neg(|X| < \aleph_0 \vee \mathbb{N} \sim X)$

The notation

- $|X| \geq \aleph_0$ denotes that X is infinite
- $|X| = \aleph_0$ denotes that X is countable
- $|X| \leq \aleph_0$ denotes that X is at most countable
- $|X| > \aleph_0$ denotes that X is uncountable

The order symbols above are not **total orders**; they are just convenient notations.

Lemma 2.2.4.

$$\forall n \in [0, \infty)_{\mathbb{N}}, \forall A \subseteq [1, n]_{\mathbb{N}}, |A| < \aleph_0$$

Proof. Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) : \forall A \subseteq [1, n]_{\mathbb{N}}, |A| < \aleph_0$$

Then

- **Base case:** Define $f : [1, 0]_{\mathbb{N}} \to [1, 0]_{\mathbb{N}}$ by

  $$\forall k \in [1, 0]_{\mathbb{N}}, f(k) = k$$

  Then

  $$\begin{gathered} [1, 0]_{\mathbb{N}} = \varnothing \\ \Downarrow \\ f : [1, 0]_{\mathbb{N}} \xrightarrow{\sim} [1, 0]_{\mathbb{N}} \\ \Downarrow \\ |[1, 0]_{\mathbb{N}}| < \aleph_0 \end{gathered}$$

  Therefore,

  $$\begin{gathered} \forall A \subseteq [1, 0]_{\mathbb{N}}, A = [1, 0]_{\mathbb{N}} \\ \Downarrow \\ \forall A \subseteq [1, 0]_{\mathbb{N}}, |A| < \aleph_0 \\ \Downarrow \\ P(0) \text{ is true} \end{gathered}$$

- **Inductive step:** Suppose
  - $P(n)$ is true
  - $A \subseteq [1, n+1]_{\mathbb{N}}$

  Then
  - If $n+1 \notin A$, then

$$\begin{array}{c} A \subseteq [1,n]_{\mathbb{N}} \\ \Downarrow \\ |A| < \aleph_0 \end{array} \qquad P(n) \text{ is true}$$

  - Otherwise, define

$$B \coloneqq A \setminus \{n+1\} \subseteq [1,n]_{\mathbb{N}}$$

    Since $P(n)$ is true,

$$\begin{array}{c} |B| < \aleph_0 \\ \Downarrow \\ \exists m \in [0,\infty)_{\mathbb{N}}, \exists f : [1,m]_{\mathbb{N}} \xrightarrow{\sim} B \end{array}$$

    Define $g : [1, m+1]_{\mathbb{N}} \to A$ by

$$\forall k \in [1,m+1]_{\mathbb{N}}, g(k) = \begin{cases} n+1, & \text{if } k = m+1 \\ f(k), & \text{otherwise} \end{cases}$$

    Then
    * **Injectivity**: Suppose
      - $k_1, k_2 \in [1, m+1]_{\mathbb{N}}$
      - $g(k_1) = g(k_2)$

      Then
      - If $k_1 = m+1$, then

$$\begin{array}{c} g(k_1) = n+1 \\ \Downarrow \\ g(k_2) = n+1 \\ \Downarrow \\ k_2 = m+1 \end{array} \qquad n+1 \notin B$$

· Otherwise,

$$\begin{array}{cr} k_1 \in [1, m]_{\mathbb{N}} & \\ \Downarrow & \\ g(k_1) = f(k_1) & \\ \Downarrow & \\ g(k_2) = f(k_1) & \\ \Downarrow & f(k_1) \neq n+1 \\ g(k_2) = f(k_2) = f(k_1) & \\ \Downarrow & f : [1, m]_{\mathbb{N}} \xrightarrow{\sim} B \\ k_1 = k_2 & \end{array}$$

∗ **Surjectivity**: Suppose $a \in A$.

· If $a = n + 1$, then

$$g(m + 1) = a$$

· Otherwise,

$$\begin{array}{cr} a \in B & \\ \Downarrow & f : [1, m]_{\mathbb{N}} \xrightarrow{\sim} B \\ \exists k \in [1, m]_{\mathbb{N}}, f(k) = a & \\ \Downarrow & \\ g(k) = f(k) = a & \end{array}$$

Therefore,

$$\begin{array}{c} g : [1, m + 1]_{\mathbb{N}} \xrightarrow{\sim} A \\ \Downarrow \\ |A| < \aleph_0 \end{array}$$

Therefore, $P(n + 1)$ is true.

By theorem 1.1.2,

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \text{ is true}$$

□

Theorem 2.2.5.

$$\forall X \forall Y, |X| < \aleph_0 \wedge Y \subseteq X \implies |Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $Y \subseteq X$

Then

$$\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

Define

$$Z := f^{-1}(Y) \subseteq [1, n]_{\mathbb{N}}$$

By lemma 2.2.4,

$$\begin{gathered} |Z| < \aleph_0 \\ \Downarrow \\ \exists m \in [0, \infty)_{\mathbb{N}}, [1, m]_{\mathbb{N}} \sim Z \end{gathered}$$

By theorem 2.1.7,

$$\begin{gathered} f|_Z^Y : Z \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| = m < \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.1}$$

□

Corollary 2.2.6.

$$\forall X \forall Y, |Y| \geq \aleph_0 \land Y \subseteq X \implies |X| \geq \aleph_0$$

Proof. Suppose

- $|Y| \geq \aleph_0$
- $Y \subseteq X$

Assume

$$|X| < \aleph_0$$

By theorem 2.2.5,

$$\begin{gathered} |Y| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|X| \geq \aleph_0$$

□

Lemma 2.2.7.

$$\forall Y, (\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \twoheadrightarrow Y) \implies |Y| < \aleph_0$$

Proof. Suppose

$$\exists n \in [0,\infty)_{\mathbb{N}}, \exists f : [1,n]_{\mathbb{N}} \twoheadrightarrow Y$$

Define

$$\forall y \in Y, N_y \coloneqq f^{-1}(\{y\}) \subseteq [1,n]_{\mathbb{N}}$$

Then

$$\begin{gathered}\forall y \in Y, N_y \neq \varnothing \\ \Downarrow \\ \forall y \in Y, \exists n_y \in N_y, n_y = \min N_y\end{gathered} \qquad \text{Theorem 1.1.4}$$

Define

$$M \coloneqq \{n_y \mid y \in Y\} \subseteq [1,n]_{\mathbb{N}}$$

Define $g : M \to Y$ by

$$g \equiv f|_M^Y$$

Then

- **Injectivity**: Suppose
  - $n_{y_1}, n_{y_2} \in M$
  - $g(n_{y_1}) = g(n_{y_2})$

  Then

$$\begin{gathered}f(n_{y_1}) = f(n_{y_2}) \\ \Downarrow \\ y_1 = y_2 \\ \Downarrow \\ N_{y_1} = N_{y_2} \\ \Downarrow \\ n_{y_1} = n_{y_2}\end{gathered}$$

- **Surjectivity**: Suppose $y \in Y$. Then

$$\begin{gathered}N_y \neq \varnothing \\ \Downarrow \\ \exists n_y \in N_y, n_y = \min N_y \\ \Downarrow \\ g(n_y) = y\end{gathered} \qquad \text{Theorem 1.1.4}$$

Therefore,

$$g : M \xrightarrow{\sim} Y$$

By lemma 2.2.4,

$$\begin{gathered}|M| < \aleph_0 \\ \Downarrow \\ \exists m \in [0,\infty)_{\mathbb{N}}, [1,m]_{\mathbb{N}} \sim M \\ \Downarrow \\ |Y| = m < \aleph_0\end{gathered} \qquad \text{Theorem 2.2.1}$$

□

Theorem 2.2.8.

$$\forall X \forall Y, |X| < \aleph_0 \wedge \exists f : X \twoheadrightarrow Y \implies |Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $\exists f : X \twoheadrightarrow Y$

Then

$$\begin{array}{cr} \exists n \in [0, \infty)_{\mathbb{N}}, \exists g : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X & \\ \Downarrow & \text{Theorem 2.1.9} \\ f \circ g : [1, n]_{\mathbb{N}} \twoheadrightarrow Y & \\ \Downarrow & \text{Lemma 2.2.7} \\ |Y| < \aleph_0 & \end{array}$$

□

Theorem 2.2.9. *Suppose X is ordered. Then*

$$0 < |X| < \aleph_0 \implies \exists m, M \in X, m = \min X \wedge M = \max X$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) : \forall X, |X| = n \implies \exists m, M \in X, m = \min X \wedge M = \max X$$

Then

- **Base case:** Suppose $X = \{a_1\}$. Then
  - $\forall x \in X, a_1 \leq x$
  - $\forall x \in X, x \leq a_1$

  Therefore, $P(1)$ is true.
- **Inductive step:** Suppose $P(n)$ is true. Suppose
  - $X = \{a_1, a_2, \ldots, a_{n+1}\}$
  - $Y = \{a_1, a_2, \ldots, a_n\} \subseteq X$

  Since $P(n)$ is true,
  - $\exists k \in [1, n]_{\mathbb{N}}, \forall y \in Y, a_k \leq y$

- $\exists K \in [1,n]_{\mathbb{N}}, \forall y \in Y, y \leq a_K$

Then

- $\forall x \in X, \min\{a_k, a_{n+1}\} \leq x$
- $\forall x \in X, x \leq \max\{a_K, a_{n+1}\}$

Therefore, $P(n+1)$ is true.

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n) \text{ is true}$$

□

Lemma 2.2.10. *Suppose* $n, m \in [0,\infty)_{\mathbb{N}}$. *Then*

(A) $(\exists f : [1,n]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}}) \implies n \leq m$

(B) $(\exists f : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}}) \implies n \geq m \geq 1 \vee n = m = 0$

(C) $\left(\exists f : [1,n]_{\mathbb{N}} \xrightarrow{\sim} [1,m]_{\mathbb{N}}\right) \implies n = m$

Proof.

(A) Define

$$\forall n \in [0,\infty)_{\mathbb{N}}, P(n) : \forall m \in [0,\infty)_{\mathbb{N}}, (\exists f_n : [1,n]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}}) \implies n \leq m$$

Then

- **Base case:**
$$\begin{gathered}\forall m \in [0,\infty)_{\mathbb{N}}, 0 \leq m \\ \Downarrow \\ P(0) \text{ is true}\end{gathered}$$
- **Inductive step:** Suppose $P(n)$ is true.
  - If $m = 0$, then
  $$\nexists f_{n+1} : [1,n+1]_{\mathbb{N}} \hookrightarrow [1,0]_{\mathbb{N}}$$
  - Otherwise, let
  $$\exists f_{n+1} : [1,n+1]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}}$$
  where
  $$f_{n+1}(n+1) = x \in [1,m]_{\mathbb{N}}$$

Define $g_m : [1, m]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$\forall y \in [1, m]_{\mathbb{N}}, g_m(y) = \begin{cases} x, & \text{if } y = m \\ m, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then

* **Injectivity**: Suppose
  - $y_1, y_2 \in [1, m]_{\mathbb{N}}$
  - $g_m(y_1) = g_m(y_2)$

  Then
  - If $y_1 = m$, then

$$\begin{gathered} g_m(y_1) = x \\ \Downarrow \\ g_m(y_2) = x \\ \Downarrow \\ y_2 = m \end{gathered}$$

  - If $y_1 = x$, then

$$\begin{gathered} g_m(y_1) = m \\ \Downarrow \\ g_m(y_2) = m \\ \Downarrow \\ y_2 = x \end{gathered}$$

  - Otherwise,

$$\begin{gathered} g_m(y_1) = y_1 \\ \Downarrow \\ g_m(y_2) = y_1 \\ \Downarrow \qquad\qquad y_1 \neq m \wedge y_1 \neq x \\ g_m(y_2) = y_2 = y_1 \end{gathered}$$

* **Surjectivity**: Suppose $k \in [1, m]_{\mathbb{N}}$. Then
  - If $k = m$, then

$$g_m(x) = k$$

  - If $k = x$, then

$$g_m(m) = k$$

  - Otherwise,

$$g_m(k) = k$$

Therefore,

* $g_m : [1,m]_{\mathbb{N}} \xrightarrow{\sim} [1,m]_{\mathbb{N}}$
* $g_m \circ f_{n+1}(n+1) = m$
* $(g_m \circ f_{n+1})([1,n]_{\mathbb{N}}) \subseteq [1,m-1]_{\mathbb{N}}$

Define $f_n : [1,n]_{\mathbb{N}} \to [1,m-1]_{\mathbb{N}}$ by

$$f_n \equiv (g_m \circ f_{n+1})|_{[1,n]_{\mathbb{N}}}^{[1,m-1]_{\mathbb{N}}}$$

By theorems 2.1.7 and 2.1.9,

$$\begin{gathered} f_n : [1,n]_{\mathbb{N}} \hookrightarrow [1,m-1]_{\mathbb{N}} \\ \Downarrow \\ n \leq m-1 \\ \Downarrow \\ n+1 \leq m \end{gathered} \qquad P(n) \text{ is true}$$

Therefore, $P(n+1)$ is true.

By theorem 1.1.2,

$$\forall n \in [0,\infty)_{\mathbb{N}}, P(n) \text{ is true}$$

(B) Define

$$\forall m \in [0,\infty)_{\mathbb{N}}, Q(m) : \forall n \in [0,\infty)_{\mathbb{N}},$$
$$(\exists f_m : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}}) \implies n \geq m \geq 1 \vee n = m = 0$$

Then

* **Base case:**
  – ($m = 0$) Suppose $n \in [0,\infty)_{\mathbb{N}}$. Consider

$$\begin{gathered} \neg(m \geq 1) \\ \Downarrow \\ n \geq m \geq 1 \vee n = m = 0 \iff n = m = 0 \iff n = 0 \end{gathered}$$

  Then

$$\begin{gathered} n \neq 0 \implies \nexists f_m : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}} \\ \Downarrow \\ Q(0) \text{ is true} \end{gathered}$$

  – ($m = 1$) Suppose $n \in [0,\infty)_{\mathbb{N}}$. Consider

$$\begin{gathered} \neg(m = 0) \\ \Downarrow \\ n \geq m \geq 1 \vee n = m = 0 \iff n \geq m \geq 1 \iff n \geq 1 \end{gathered}$$

Then

$$\begin{gathered} n < 1 \implies \nexists f_m : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}} \\ \Downarrow \\ Q(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Suppose $Q(m)$ is true.
  - If $n = 0$, then

$$\nexists f_{m+1} : [1,0]_{\mathbb{N}} \twoheadrightarrow [1,m+1]_{\mathbb{N}}$$

  - Otherwise, let

$$\exists f_{m+1} : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m+1]_{\mathbb{N}}$$

where

$$f_{m+1}(n) = x \in [1,m+1]_{\mathbb{N}}$$

Define $g_{m+1} : [1,m+1]_{\mathbb{N}} \to [1,m+1]_{\mathbb{N}}$ by

$$\forall y \in [1,m+1]_{\mathbb{N}}, g_{m+1}(y) = \begin{cases} x, & \text{if } y = m+1 \\ m+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then

* **Injectivity**: Suppose
  - $y_1, y_2 \in [1,m+1]_{\mathbb{N}}$
  - $g_{m+1}(y_1) = g_{m+1}(y_2)$

  Then
  - If $y_1 = m+1$, then

$$\begin{gathered} g_{m+1}(y_1) = x \\ \Downarrow \\ g_{m+1}(y_2) = x \\ \Downarrow \\ y_2 = m+1 \end{gathered}$$

  - If $y_1 = x$, then

$$\begin{gathered} g_{m+1}(y_1) = m+1 \\ \Downarrow \\ g_{m+1}(y_2) = m+1 \\ \Downarrow \\ y_2 = x \end{gathered}$$

· Otherwise,

$$\begin{array}{c} g_{m+1}(y_1) = y_1 \\ \Downarrow \\ g_{m+1}(y_2) = y_1 \\ \Downarrow \\ g_{m+1}(y_2) = y_2 = y_1 \end{array} \qquad y_1 \neq m+1 \wedge y_1 \neq x$$

∗ **Surjectivity**: Suppose $k \in [1, m+1]_{\mathbb{N}}$. Then

· If $k = m+1$, then

$$g_{m+1}(x) = k$$

· If $k = x$, then

$$g_{m+1}(m+1) = k$$

· Otherwise,

$$g_{m+1}(k) = k$$

Therefore,

∗ $g_{m+1} : [1, m+1]_{\mathbb{N}} \xrightarrow{\sim} [1, m+1]_{\mathbb{N}}$

∗ $(g_{m+1} \circ f_{m+1})(n) = m+1$

Define

$$K_{m+1} \coloneqq (g_{m+1} \circ f_{m+1})^{-1}(\{m+1\}) \subseteq [1, n]_{\mathbb{N}}$$

Then

$$(g_{m+1} \circ f_{m+1})([1, n-1]_{\mathbb{N}} \setminus K_{m+1}) = [1, m]_{\mathbb{N}}$$

Define $f_m : [1, n-1]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$\forall y \in [1, n-1]_{\mathbb{N}}, f_m(y) = \begin{cases} m, & \text{if } y \in K_{m+1} \\ (g_{m+1} \circ f_{m+1})(y), & \text{otherwise} \end{cases}$$

By theorems 2.1.7 and 2.1.9,

$$\begin{array}{c} (g_{m+1} \circ f_{m+1})|_{[1,n-1]_{\mathbb{N}} \setminus K_{m+1}}^{[1,m]_{\mathbb{N}}} : [1, n-1]_{\mathbb{N}} \setminus K_{m+1} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ f_m : [1, n-1]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n-1 \geq m \\ \Downarrow \\ n \geq m+1 \geq 1 \end{array} \qquad Q(m) \text{ is true}$$

Therefore, $Q(m+1)$ is true.

By theorem 1.1.2,

$$\forall m \in [0,\infty)_{\mathbb{N}}, Q(m) \text{ is true}$$

(C) follows from **(A)** and **(B)**.

□

Lemma 2.2.11. *Suppose* $n, m \in [0,\infty)_{\mathbb{N}}$. *Then*

(A) $n \leq m \implies \exists f : [1,n]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}}$

(B) $n \geq m \geq 1 \lor n = m = 0 \implies \exists f : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}}$

(C) $n = m \implies \exists f : [1,n]_{\mathbb{N}} \xrightarrow{\sim} [1,m]_{\mathbb{N}}$

This lemma is the converse of lemma 2.2.10.

Proof.

(A) Suppose $n \leq m$. Define $f : [1,n]_{\mathbb{N}} \to [1,m]_{\mathbb{N}}$ by

$$\forall i \in [1,n]_{\mathbb{N}}, f(i) = i$$

Suppose

- $i, j \in [1,n]_{\mathbb{N}}$
- $f(i) = f(j)$

Then

$$\begin{gathered} f(i) = f(j) \\ \Downarrow \\ i = j \end{gathered}$$

Therefore,

$$f : [1,n]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}}$$

(B)

- Suppose $n = m = 0$. Define $f : [1,n]_{\mathbb{N}} \to [1,m]_{\mathbb{N}}$ by

$$\forall i \in [1,n]_{\mathbb{N}}, f(i) = i$$

Then

$$f : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}}$$

- Suppose $n \geq m \geq 1$. Define $f : [1, n]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$\forall i \in [1, n]_{\mathbb{N}}, f(i) = \begin{cases} i, & \text{if } i < m \\ m, & \text{if } i \geq m \end{cases}$$

  Suppose $j \in [1, m]_{\mathbb{N}}$. Then
  - If $j = m$, then

$$f(n) = j$$

  - Otherwise,

$$f(j) = j$$

  Therefore,

$$f : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}$$

(C) Suppose $n = m$. Define $f : [1, n]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$\forall i \in [1, n]_{\mathbb{N}}, f(i) = i$$

Then

- **Injectivity**: Suppose
  - $i, j \in [1, n]_{\mathbb{N}}$
  - $f(i) = f(j)$

  Then

$$\begin{gathered} f(i) = f(j) \\ \Downarrow \\ i = j \end{gathered}$$

  Therefore,

$$f : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}$$

- **Surjectivity**: Suppose $j \in [1, m]_{\mathbb{N}}$. Then

$$f(j) = j$$

  Therefore,

$$f : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}$$

Therefore,

$$f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}$$

□

Theorem 2.2.12. *Suppose*

- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

(A) $|X| \leq |Y| \iff \exists f : X \hookrightarrow Y$

(B) $|X| \geq |Y| > 0 \vee |X| = |Y| = 0 \iff \exists f : X \twoheadrightarrow Y$

(C) $|X| = |Y| \iff \exists f : X \xrightarrow{\sim} Y$

The contraposition of ***(A,*** $\Longleftarrow$ ***)*** *is called the* ***pigeonhole principle****.*

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

- $\exists n \in [0, \infty)_{\mathbb{N}}, \exists f_X : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists m \in [0, \infty)_{\mathbb{N}}, \exists f_Y : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$

Therefore,

- **(** $\Longleftarrow$ **)** Suppose $f : X \to Y$. Define $F : [1, n]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$F \equiv (f_Y)^{-1} \circ f \circ f_X$$

By theorems 2.1.5 and 2.1.9,

  - $f : X \hookrightarrow Y \implies F : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}$
  - $f : X \twoheadrightarrow Y \implies F : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}$

This implies

(A)

$$\begin{array}{c} \exists f : X \hookrightarrow Y \\ \Downarrow \\ \exists F : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n \leq m \end{array}$$

Lemma 2.2.10

(B)

$$\begin{array}{c} \exists f : X \twoheadrightarrow Y \\ \Downarrow \\ \exists F : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n \geq m \geq 1 \vee n = m = 0 \end{array}$$

Lemma 2.2.10

(C)

$$\begin{array}{c}\exists f : X \xrightarrow{\sim} Y \\ \Downarrow \\ \exists F : [1,n]_{\mathbb{N}} \xrightarrow{\sim} [1,m]_{\mathbb{N}} \\ \Downarrow \\ n = m\end{array} \qquad \text{Lemma 2.2.10}$$

- ($\Longrightarrow$) Suppose $G : [1,n]_{\mathbb{N}} \to [1,m]_{\mathbb{N}}$. Define $f : X \to Y$ by

$$f \equiv f_Y \circ G \circ (f_X)^{-1}$$

By theorems 2.1.5 and 2.1.9,

 - $G : [1,n]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}} \implies f : X \hookrightarrow Y$
 - $G : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}} \implies f : X \twoheadrightarrow Y$

This implies

(A)

$$\begin{array}{c}n \le m \\ \Downarrow \\ \exists G : [1,n]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \hookrightarrow Y\end{array} \qquad \text{Lemma 2.2.11}$$

(B)

$$\begin{array}{c}n \ge m \ge 1 \vee n = m = 0 \\ \Downarrow \\ \exists G : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \twoheadrightarrow Y\end{array} \qquad \text{Lemma 2.2.11}$$

(C)

$$\begin{array}{c}n = m \\ \Downarrow \\ \exists G : [1,n]_{\mathbb{N}} \xrightarrow{\sim} [1,m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \xrightarrow{\sim} Y\end{array} \qquad \text{Lemma 2.2.11}$$

□

Lemma 2.2.13.

$$\forall X \forall Y, X \cap Y = \varnothing \wedge |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Suppose

- $X \cap Y = \varnothing$
- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

- $\exists n \in [0,\infty)_{\mathbb{N}}, \exists f : [1,n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists m \in [0,\infty)_{\mathbb{N}}, \exists g : [1,m]_{\mathbb{N}} \xrightarrow{\sim} Y$

Define $h : [1,m+n]_{\mathbb{N}} \to X \cup Y$ by

$$\forall i \in [1,m+n]_{\mathbb{N}}, h(i) = \begin{cases} f(i), & \text{if } i \leq n \\ g(i-n), & \text{otherwise} \end{cases}$$

Then

- **Injectivity**: Suppose
  - $i, j \in [1,m+n]_{\mathbb{N}}$
  - $h(i) = h(j)$

  Then

  - If $i \leq n$, then

$$\begin{array}{cr} h(i) = f(i) & \\ \Downarrow & \\ f(i) = h(j) \in X & \\ \Downarrow & X \cap Y = \varnothing \\ j \leq n & \\ \Downarrow & \\ h(j) = f(j) & \\ \Downarrow & \\ f(i) = f(j) & \\ \Downarrow & f : [1,n]_{\mathbb{N}} \xrightarrow{\sim} X \\ i = j & \end{array}$$

- Otherwise,

$$\begin{array}{cr} h(i) = g(i-n) & \\ \Downarrow & \\ g(i-n) = h(j) \in Y & \\ \Downarrow & X \cap Y = \varnothing \\ j > n & \\ \Downarrow & \\ h(j) = g(j-n) & \\ \Downarrow & \\ g(i-n) = g(j-n) & \\ \Downarrow & g : [1,m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ i-n = j-n & \\ \Downarrow & \\ i = j & \end{array}$$

- **Surjectivity**: Suppose $k \in X \cup Y$. Then
  - If $k \in X$, then

$$\begin{array}{cr} k \in X & \\ \Downarrow & f : [1,n]_{\mathbb{N}} \xrightarrow{\sim} X \\ \exists i \in [1,n]_{\mathbb{N}}, f(i) = k & \\ \Downarrow & \\ h(i) = k & \end{array}$$

  - Otherwise,

$$\begin{array}{cr} k \in Y & \\ \Downarrow & g : [1,m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ \exists j \in [1,m]_{\mathbb{N}}, g(j) = k & \\ \Downarrow & \\ h(j+n) = k & \end{array}$$

Therefore,

$$\begin{array}{c} h : [1,m+n]_{\mathbb{N}} \xrightarrow{\sim} X \cup Y \\ \Downarrow \\ |X \cup Y| = m+n < \aleph_0 \end{array}$$

□

Lemma 2.2.14.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$

- $|Y| < \aleph_0$

Define

- $A := X \setminus Y \subseteq X$
- $B := Y \setminus X \subseteq Y$
- $C := X \cap Y \subseteq X$

By theorem 2.2.5,

- $|A| < \aleph_0$
- $|B| < \aleph_0$
- $|C| < \aleph_0$

Also,

- $A \cap B = \varnothing$
- $B \cap C = \varnothing$
- $C \cap A = \varnothing$

Consider

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$
- $X \cup Y = (X \setminus Y) \cup (Y \setminus X) \cup (X \cap Y) = A \cup B \cup C$

By lemma 2.2.13,

$$\begin{gathered} |A \cup B| < \aleph_0 \\ \Downarrow \\ |(A \cup B) \cup C| < \aleph_0 \\ \Downarrow \\ |X \cup Y| < \aleph_0 \end{gathered}$$

□

Theorem 2.2.15.

$$\forall n \in \mathbb{N}, \forall X_1, X_2, \ldots, X_n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \bigcup_{k=1}^{n} X_k \right| < \aleph_0$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) : (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \bigcup_{k=1}^{n} X_k \right| < \aleph_0$$

Then

- **Base case:** Suppose
  - $|X_1| < \aleph_0$

  Then

$$\begin{gathered} |X_1| < \aleph_0 \\ \Downarrow \\ \left| \bigcup_{k=1}^{1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Suppose $P(n)$ is true. Suppose
  - $\forall k \in [1, n+1]_{\mathbb{N}}, |X_k| < \aleph_0$

  Since $P(n)$ is true,

$$\left| \bigcup_{k=1}^{n} X_k \right| < \aleph_0$$

  Then

$$\begin{gathered} \bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^{n} X_k \right) \cup X_{n+1} \\ \Downarrow \qquad \text{Lemma 2.2.14} \\ \left| \bigcup_{k=1}^{n+1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n) \text{ is true}$$

□

Lemma 2.2.16.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \times Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

- $\exists n \in [0,\infty)_{\mathbb{N}}, \exists f : [1,n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists m \in [0,\infty)_{\mathbb{N}}, \exists g : [1,m]_{\mathbb{N}} \xrightarrow{\sim} Y$

Therefore,

- If $n = 0$ or $m = 0$, then
$$\begin{gathered} X \times Y = \varnothing \\ \Downarrow \\ |X \times Y| = 0 < \aleph_0 \end{gathered}$$
- Otherwise, define $h : [1,nm]_{\mathbb{N}} \to X \times Y$ as
$$\forall k \in [1,nm]_{\mathbb{N}}, h(k) = \left(f\left(\left\lceil \frac{k}{m} \right\rceil\right), g\left(k - m\left(\left\lceil \frac{k}{m} \right\rceil - 1\right)\right)\right)$$
Suppose
  - $k \in [1,nm]_{\mathbb{N}}$
  - $j_k = \left\lceil \dfrac{k}{m} \right\rceil \in \mathbb{Z}$

  Then
  - Since $k \in [1,nm]_{\mathbb{N}}$,
$$\begin{gathered} 1 \leq k \leq nm \\ \Downarrow \\ 1 \leq \left\lceil \frac{k}{m} \right\rceil \leq n \\ \Downarrow \\ j_k \in [1,n]_{\mathbb{N}} \end{gathered}$$
  - By definition 1.6.6,
$$\begin{gathered} j_k = \min\left\{ i \in \mathbb{Z} \;\middle|\; i \geq \frac{k}{m} \right\} \\ \Downarrow \\ j_k - 1 < \frac{k}{m} \leq j_k \\ \Downarrow \\ 0 < k - m(j_k - 1) \leq m \\ \Downarrow \\ k - m(j_k - 1) \in [1,m]_{\mathbb{N}} \end{gathered}$$

Therefore, h is well-defined. Then

- **Injectivity**: Suppose
  * $k_1, k_2 \in [1, nm]_{\mathbb{N}}$
  * $h(k_1) = h(k_2)$

  Then

$$\begin{aligned} f(j_{k_1}) &= f(j_{k_2}) \\ &\Downarrow && f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X \\ j_{k_1} &= j_{k_2} \end{aligned}$$

  Therefore,

$$\begin{aligned} g(k_1 - m(j_{k_1} - 1)) &= g(k_2 - m(j_{k_2} - 1)) \\ &\Downarrow && g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ k_1 - m(j_{k_1} - 1) &= k_2 - m(j_{k_2} - 1) \\ &\Downarrow && j_{k_1} = j_{k_2} \\ k_1 &= k_2 \end{aligned}$$

- **Surjectivity**: Suppose $(x, y) \in X \times Y$. Then
  * Since $(x, y) \in X \times Y$,

$$\begin{aligned} &x \in X \\ &\Downarrow && f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X \\ &\exists j \in [1, n]_{\mathbb{N}}, f(j) = x \end{aligned}$$

  * Since $(x, y) \in X \times Y$,

$$\begin{aligned} &y \in Y \\ &\Downarrow && g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ &\exists i \in [1, m]_{\mathbb{N}}, g(i) = y \end{aligned}$$

  Let

$$k \coloneqq m(j - 1) + i \in [1, nm]_{\mathbb{N}}$$

  Then

$$h(k) = (f(j), g(i)) = (x, y)$$

Therefore,

$$\begin{aligned} &h : [1, nm]_{\mathbb{N}} \xrightarrow{\sim} X \times Y \\ &\Downarrow \\ &|X \times Y| = nm < \aleph_0 \end{aligned}$$

□

Theorem 2.2.17.

$$\forall n \in \mathbb{N}, \forall X_1, X_2, \ldots, X_n, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^{n} X_k \right| < \aleph_0$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) : (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^{n} X_k \right| < \aleph_0$$

Then

- **Base case:** Suppose
  - $|X_1| < \aleph_0$

  Then

$$\begin{gathered} |X_1| < \aleph_0 \\ \Downarrow \\ \left| \prod_{k=1}^{1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Suppose $P(n)$ is true. Suppose
  - $\forall k \in [1, n+1]_{\mathbb{N}}, |X_k| < \aleph_0$

  Since $P(n)$ is true,

$$\left| \prod_{k=1}^{n} X_k \right| < \aleph_0$$

  Then

$$\begin{gathered} \prod_{k=1}^{n+1} X_k \sim \left(\prod_{k=1}^{n} X_k \right) \times X_{n+1} \\ \Downarrow \qquad \text{Lemma 2.2.16} \\ \left| \prod_{k=1}^{n+1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n) \text{ is true}$$

□

Theorem 2.2.18.

$$\forall X, \nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

This theorem is called ***Cantor's theorem****.*

Proof. Suppose

$$\exists f : X \twoheadrightarrow \mathcal{P}(X)$$

Define

$$S := \{x \in X \mid x \notin f(x)\} \subseteq X$$

Then

$$\begin{gathered} S \subseteq X \\ \Downarrow \\ S \in \mathcal{P}(X) \\ \Downarrow \\ \exists x \in X, f(x) = S \end{gathered}$$

Consider

- If $x \in S$, then

$$\begin{gathered} x \notin f(x) = S \\ \Downarrow \\ \bot \end{gathered}$$

- If $x \notin S$, then

$$\begin{gathered} x \in f(x) = S \\ \Downarrow \\ \bot \end{gathered}$$

Therefore,

$$\nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

□

2.3 Countable and Uncountable Sets

Theorem 2.3.1.

$$\forall X \forall Y, |X| \geq \aleph_0 \land |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $|Y| < \aleph_0$

Assume

$$\begin{gathered} |X \setminus Y| < \aleph_0 \\ \Downarrow \\ |(X \setminus Y) \cup Y| < \aleph_0 \end{gathered} \qquad \text{Lemma 2.2.14}$$

Then

$$\begin{gathered} X = (X \setminus Y) \cup (X \cap Y) \\ \Downarrow \\ X \subseteq (X \setminus Y) \cup Y \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \bot \end{gathered} \qquad \text{Theorem 2.2.5}$$

Therefore,

$$|X \setminus Y| \geq \aleph_0$$

□

Theorem 2.3.2.

$$\forall X \forall Y, |X| \geq \aleph_0 \land \exists f : X \hookrightarrow Y \implies |Y| \geq \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $\exists f : X \hookrightarrow Y$

Assume

$$|Y| < \aleph_0$$

Define $h : f(X) \to X$ by

$$h \equiv \left(f|^{f(X)}\right)^{-1}$$

Define $g : Y \to X$ by

$$\forall y \in Y, g(y) = \begin{cases} h(y), & \text{if } y \in f(X) \\ h(y_0), & \text{otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. By theorem 2.1.5,

$$\begin{gathered} h : f(X) \xrightarrow{\sim} X \\ \Downarrow \\ g : Y \twoheadrightarrow X \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \bot \end{gathered} \qquad \text{Theorem 2.2.8}$$

Therefore,

$$|Y| \geq \aleph_0$$

□

Lemma 2.3.3.

$$|\mathbb{N}| \geq \aleph_0$$

Proof. Assume

$$|\mathbb{N}| < \aleph_0$$

Then

$$\begin{gathered} 1 \in \mathbb{N} \\ \Downarrow \\ 0 < |\mathbb{N}| \\ \Downarrow \qquad \text{Theorem 2.2.9} \\ \exists M \in \mathbb{N}, M = \max(\mathbb{N}) \end{gathered}$$

Then

$$\begin{gathered} M < M + 1 \\ \Downarrow \\ M + 1 \notin \mathbb{N} \\ \Downarrow \qquad \text{Axiom 1.1.1} \\ \bot \end{gathered}$$

Therefore,

$$|\mathbb{N}| \geq \aleph_0$$

□

Theorem 2.3.4.

$$\forall X, |X| = \aleph_0 \implies |X| \geq \aleph_0$$

Proof. Suppose $|X| = \aleph_0$. Then

$$\begin{gathered} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \qquad \text{Theorem 2.3.2 and lemma 2.3.3} \\ |X| \geq \aleph_0 \end{gathered}$$

□

Theorem 2.3.5.

$$|\mathbb{Z}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ by

$$\forall n \in \mathbb{N}, f(n) = \begin{cases} -\dfrac{n}{2}, & \text{if } n \text{ is even} \\ \dfrac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

Then

- **Injectivity**: Suppose
  - $n_1, n_2 \in \mathbb{N}$
  - $f(n_1) = f(n_2)$

  Then
  - n is even $\implies f(n) = -\dfrac{n}{2} < 0$
  - n is odd $\implies f(n) = \dfrac{n-1}{2} \geq 0$

  Therefore, n_1 and n_2 are both even or both odd.
  - If both n_1 and n_2 are even, then
  $$-\frac{n_1}{2} = -\frac{n_2}{2}$$
  $$\Downarrow$$
  $$n_1 = n_2$$
  - Otherwise,
  $$\frac{n_1 - 1}{2} = \frac{n_2 - 1}{2}$$
  $$\Downarrow$$
  $$n_1 = n_2$$
- **Surjectivity**: Suppose $k \in \mathbb{Z}$.
  - If $k \geq 0$, then
  $$2k + 1 \in \mathbb{N}$$
  $$\Downarrow$$
  $$f(2k+1) = \frac{(2k+1) - 1}{2} = \frac{2k}{2} = k$$
  - If $k < 0$, then
  $$-2k \in \mathbb{N}$$
  $$\Downarrow$$
  $$f(-2k) = -\frac{-2k}{2} = k$$

Therefore,

$$\begin{gathered} f : \mathbb{N} \xrightarrow{\sim} \mathbb{Z} \\ \Downarrow \\ |\mathbb{Z}| = \aleph_0 \end{gathered}$$

□

Lemma 2.3.6.

$$\forall X \subseteq \mathbb{N}, |X| \geq \aleph_0 \implies |X| = \aleph_0$$

Proof. Suppose

- $X \subseteq \mathbb{N}$
- $|X| \geq \aleph_0$

Define

- $(X_n)_{n=1}^{\infty}$ in $\mathcal{P}(X)$
- $f : \mathbb{N} \to X$

as

- $X_1 \coloneqq X$
- $\forall n \in \mathbb{N}, X_{n+1} \coloneqq X_n \setminus \{f(n)\}$
- $\forall n \in \mathbb{N}, f(n) \coloneqq \min X_n$

By theorem 2.3.1,

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| \geq \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, X_n \neq \varnothing \\ \Downarrow \qquad \text{Theorem 1.1.4} \\ \forall n \in \mathbb{N}, \exists f(n) \in X_n, f(n) = \min X_n \end{gathered}$$

Therefore, f is well-defined. Then

- **Injectivity**: Suppose
  - $n_1, n_2 \in \mathbb{N}$
  - $n_1 \neq n_2$ (*WLOG* $n_1 < n_2$)

  Then

  $$\begin{gathered} X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\} \\ \Downarrow \\ f(n_2) \neq f(n_1) \end{gathered}$$

  Therefore,

  $$f : \mathbb{N} \hookrightarrow X$$

- **Surjectivity**: Suppose $x \in X$. Define

$$S := \{n \in X \mid n < x\} \subseteq [1, x-1]_{\mathbb{N}}$$

By lemma 2.2.4,

$$\begin{gathered} |S| < \aleph_0 \\ \Downarrow \\ \exists k \in [0, \infty)_{\mathbb{N}}, |S| = k \end{gathered}$$

  - If $k = 0$, then
    * $S = \varnothing$
    * $x = \min X_1 = f(1)$
  - Otherwise, let
    * $S = \{s_1, s_2, \ldots, s_k\}$
    * $s_1 < s_2 < \cdots < s_k$

    Then

$$\begin{aligned} \min X_1 &= \min S \\ \min X_2 &= \min S \setminus \{s_1\} \\ &\vdots \\ \min X_k &= \min S \setminus \{s_1, s_2, \ldots, s_{k-1}\} \end{aligned} \qquad \begin{aligned} X_2 &= X \setminus \{s_1\} \\ X_3 &= X \setminus \{s_1, s_2\} \\ &\vdots \\ X_{k+1} &= X \setminus S \end{aligned}$$

    Therefore,

$$\begin{gathered} \forall n \in X_{k+1}, n \geq x \\ \Downarrow \\ x = \min X_{k+1} \\ \Downarrow \\ f(k+1) = x \end{gathered}$$

  Therefore,

$$f : \mathbb{N} \twoheadrightarrow X$$

Therefore,

$$\begin{gathered} f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ |X| = \aleph_0 \end{gathered}$$

□

Theorem 2.3.7.

$$\forall X \forall Y, Y \subseteq X \wedge |Y| \geq \aleph_0 \wedge |X| = \aleph_0 \implies |Y| = \aleph_0$$

Proof. Suppose

- $Y \subseteq X$
- $|X| = \aleph_0$
- $|Y| \geq \aleph_0$

Then

$$\begin{gathered} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ f|^{Y}_{f^{-1}(Y)} : f^{-1}(Y) \xrightarrow{\sim} Y \end{gathered} \qquad \text{Theorem 2.1.7}$$

Assume $\left|f^{-1}(Y)\right| < \aleph_0$. By theorem 2.2.8,

$$\begin{gathered} |Y| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\left|f^{-1}(Y)\right| \geq \aleph_0$$

By lemma 2.3.6,

$$\begin{gathered} \left|f^{-1}(Y)\right| = \aleph_0 \\ \Downarrow \\ |Y| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.1}$$

□

Theorem 2.3.8.

$$\forall X, |X| \geq \aleph_0 \wedge (\exists f : \mathbb{N} \twoheadrightarrow X) \implies |X| = \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $\exists f : \mathbb{N} \twoheadrightarrow X$

Define

$$\forall k \in X, S_k := f^{-1}(\{k\}) \neq \varnothing$$

By theorem 1.1.4,

$$\forall k \in X, \exists m_k \in S_k, m_k = \min S_k$$

Define

$$T := \{m_k \mid k \in X\} \subseteq \mathbb{N}$$

Define $g : T \to X$ by

$$g \equiv f|^{X}_{T}$$

Then

- **Injectivity**: Suppose
  - $m_{k_1}, m_{k_2} \in T$
  - $g(m_{k_1}) = g(m_{k_2})$

  Then
$$\begin{gathered} f(m_{k_1}) = f(m_{k_2}) \\ \Downarrow \\ k_1 = k_2 \\ \Downarrow \\ S_{k_1} = S_{k_2} \\ \Downarrow \\ m_{k_1} = m_{k_2} \end{gathered}$$

- **Surjectivity**: Suppose $k \in X$. By theorem 1.1.4,
$$\begin{gathered} \exists m_k \in S_k, m_k = \min S_k \\ \Downarrow \\ g(m_k) = k \end{gathered}$$

Therefore,
$$\begin{aligned} & g : T \xrightarrow{\sim} X \\ & \Downarrow && \text{Theorems 2.2.1 and 2.3.2} \\ & |T| \geq \aleph_0 \\ & \Downarrow && \text{Lemma 2.3.6} \\ & |T| = \aleph_0 \\ & \Downarrow && \text{Theorem 2.2.1} \\ & |X| = \aleph_0 \end{aligned}$$

□

Theorem 2.3.9.
$$\forall (X_n)_{n=1}^{\infty}, (\forall n \in \mathbb{N}, |X_n| < \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| \leq \aleph_0$$

Proof. Suppose
$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| < \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists m_n \in [0, \infty)_{\mathbb{N}}, \exists g_n : [1, m_n]_{\mathbb{N}} \xrightarrow{\sim} X_n \end{gathered}$$

- If $\left| \bigcup_{n=1}^{\infty} X_n \right| < \aleph_0$, then the proof is complete.

- Otherwise, define $h : \mathbb{N} \to \bigcup_{n=1}^{\infty} X_n$ as
  - $\forall m \in [1, m_1]_{\mathbb{N}}, h(m) = g_1(m)$
  - $\forall n \in [2, \infty)_{\mathbb{N}}, \forall m \in [1, m_n]_{\mathbb{N}}, h\left(\sum_{k=1}^{n-1} m_k + m\right) = g_n(m)$

  Suppose $x \in \bigcup_{n=1}^{\infty} X_n$. Then

  $$\begin{array}{c} \exists n \in \mathbb{N}, x \in X_n \\ \Downarrow \\ \exists m \in [1, m_n]_{\mathbb{N}}, g_n(m) = x \end{array} \qquad g_n : [1, m_n]_{\mathbb{N}} \xrightarrow{\sim} X_n$$

  Consider

  - If $n = 1$, then
  $$h(m) = x$$
  - Otherwise,
  $$h\left(\sum_{k=1}^{n-1} m_k + m\right) = x$$

  Therefore,

  $$\begin{array}{c} h : \mathbb{N} \twoheadrightarrow \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ \left|\bigcup_{n=1}^{\infty} X_n\right| = \aleph_0 \end{array} \qquad \text{Theorem 2.3.8}$$

□

Remark. Strictly speaking, theorem 2.3.9 requires the **axiom of choice** to prove. In this textbook, the **axiom of choice** is not discussed, so we will just assume that the proof is valid. Similar issues arise for theorems 2.3.10 and 2.3.12.

Theorem 2.3.10.

$$\forall (X_n)_{n=1}^{\infty}, (\forall n \in \mathbb{N}, |X_n| = \aleph_0) \implies \left|\bigcup_{n=1}^{\infty} X_n\right| = \aleph_0$$

Proof. Suppose

$$\begin{gathered}\forall n \in \mathbb{N}, |X_n| = \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f_n : \mathbb{N} \xrightarrow{\sim} X_n\end{gathered}$$

Assume

$$\left| \bigcup_{n=1}^{\infty} X_n \right| < \aleph_0$$

Then

$$\begin{gathered}X_1 \subseteq \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ |X_1| < \aleph_0 \\ \Downarrow \\ \perp\end{gathered}$$

Theorem 2.2.5

Theorem 2.3.4

Therefore,

$$\left| \bigcup_{n=1}^{\infty} X_n \right| \geq \aleph_0$$

Define

$$\forall n \in \mathbb{N}, T_n := \left\{ k \in \mathbb{N} \;\middle|\; n \leq \frac{k(k+1)}{2} \right\} \subseteq \mathbb{N}$$

Then

$$\begin{gathered}\forall n \in \mathbb{N}, 2n \in T_n \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists t_n \in T_n, t_n = \min T_n\end{gathered}$$

Theorem 1.1.4

Define

$$\forall n \in \mathbb{N}, s_n := n - \frac{t_n(t_n - 1)}{2}$$

Then

$$\begin{gathered}t_n - 1 \notin T_n \\ \Downarrow \\ n > \frac{t_n(t_n - 1)}{2} \\ \Downarrow \\ s_n \in \mathbb{N}\end{gathered}$$

Also,

$$\begin{gathered}t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \\ \Downarrow \\ t_n - s_n + 1 \geq 1 \\ \Downarrow \\ t_n - s_n + 1 \in \mathbb{N}\end{gathered}$$

$t_n \in T_n$

Define $g : \mathbb{N} \to \bigcup_{n=1}^{\infty} X_n$ by

$$\forall n \in \mathbb{N}, g(n) = f_{s_n}(t_n - s_n + 1)$$

Suppose $x \in \bigcup_{n=1}^{\infty} X_n$. Then

$$\begin{gathered} \exists m \in \mathbb{N}, x \in X_m \\ \Downarrow \\ \exists l \in \mathbb{N}, f_m(l) = x \end{gathered} \qquad f_m : \mathbb{N} \xrightarrow{\sim} X_m$$

Define

$$k := \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}$$

Consider

- $k - \dfrac{(l+m-2)(l+m-1)}{2} = m > 0$
- $k - \dfrac{(l+m-1)(l+m)}{2} = 1 - l \leq 0$

Then

$$\begin{gathered} l+m-2 \notin T_k \wedge l+m-1 \in T_k \\ \Downarrow \\ t_k = l+m-1 \\ \Downarrow \\ s_k = m \\ \Downarrow \\ g(k) = f_m(l) = x \end{gathered}$$

Therefore,

$$\begin{gathered} g : \mathbb{N} \twoheadrightarrow \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.3.8}$$

□

Corollary 2.3.11.

$$\forall n \in \mathbb{N}, \forall X_1, X_2, \ldots, X_n, (\forall k \in [1,n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left| \bigcup_{k=1}^{n} X_k \right| = \aleph_0$$

Proof. Suppose

$$\forall k \in [1,n]_{\mathbb{N}}, |X_k| = \aleph_0$$

Define

$$\forall k \in [n+1,\infty)_{\mathbb{N}}, X_k := X_n$$

Then

$$\begin{gathered}\forall n \in \mathbb{N}, |X_n| = \aleph_0 \\ \Downarrow \\ \left|\bigcup_{n=1}^{\infty} X_n\right| = \aleph_0 \\ \Downarrow \\ \left|\bigcup_{k=1}^{n} X_k\right| = \aleph_0\end{gathered}$$

Theorem 2.3.10

$\square$

Theorem 2.3.12.

$$\forall (X_n)_{n=1}^{\infty}, (\forall n \in \mathbb{N}, |X_n| \leq \aleph_0) \implies \left|\bigcup_{n=1}^{\infty} X_n\right| \leq \aleph_0$$

Proof. Suppose

$$\forall n \in \mathbb{N}, |X_n| \leq \aleph_0$$

Define

$$\forall n \in \mathbb{N}, Y_n := \begin{cases} X_n, & \text{if } |X_n| = \aleph_0 \\ X_n \cup \{y_n\}, & \text{otherwise} \end{cases}$$

Then

$$\begin{gathered}\forall n \in \mathbb{N}, Y_n = X_n \vee Y_n = X_n \cup \{y_n\} \\ \Downarrow \\ \forall n \in \mathbb{N}, X_n = Y_n - Y_n \vee X_n = Y_n - \{y_n\} \\ \Downarrow \\ \forall n \in \mathbb{N}, |X_n| \leq |Y_n| \wedge |Y_n| = \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, |X_n| \leq |Y_n| \wedge \left|\bigcup_{n=1}^{\infty} Y_n\right| = \aleph_0 \\ \Downarrow \\ \left|\bigcup_{n=1}^{\infty} X_n\right| \leq \left|\bigcup_{n=1}^{\infty} Y_n\right| = \aleph_0\end{gathered}$$

Theorem 2.3.10

$\square$

Lemma 2.3.13.

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$\forall m, n \in \mathbb{N}, f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

Then

- **Injectivity**: Suppose
  - $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$
  - $(m_1, n_1) \neq (m_2, n_2)$

  Then

  - If $m_1 + n_1 \neq m_2 + n_2$ (*WLOG* $m_1 + n_1 < m_2 + n_2$), then

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1+n_1-1)(m_1+n_1-2)}{2} + m_1 \\ &\leq \frac{(m_1+n_1-1)(m_1+n_1-2)}{2} + (m_1+n_1-1) \\ &= \frac{(m_1+n_1)(m_1+n_1-1)}{2} \\ &\leq \frac{(m_2+n_2-1)(m_2+n_2-2)}{2} \\ &< \frac{(m_2+n_2-1)(m_2+n_2-2)}{2} + m_2 = f(m_2, n_2) \end{aligned}$$

  - Otherwise,

$$\begin{gathered} m_1 \neq m_2 \\ \Downarrow \\ f(m_1, n_1) \neq f(m_2, n_2) \end{gathered}$$

- **Surjectivity**: Suppose $k \in \mathbb{N}$. Define t_k, s_k as in theorem 2.3.10. Then

$$f(s_k, t_k - s_k + 1) = k$$

Therefore,

$$\begin{gathered} f : \mathbb{N} \times \mathbb{N} \xrightarrow{\sim} \mathbb{N} \\ \Downarrow \\ |\mathbb{N} \times \mathbb{N}| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.1}$$

□

Theorem 2.3.14.

$$\forall X \forall Y, |X| = \aleph_0 \wedge |Y| = \aleph_0 \implies |X \times Y| = \aleph_0$$

Proof. Suppose

- $|X| = \aleph_0$
- $|Y| = \aleph_0$

Then

- $\exists f : \mathbb{N} \xrightarrow{\sim} X$
- $\exists g : \mathbb{N} \xrightarrow{\sim} Y$

Define $h : \mathbb{N} \times \mathbb{N} \to X \times Y$ by

$$\forall m, n \in \mathbb{N}, h(m, n) = (f(m), g(n))$$

Then

- **Injectivity**: Suppose
  - $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$
  - $h(m_1, n_1) = h(m_2, n_2)$

  Then

$$\begin{gathered} f(m_1) = f(m_2) \wedge g(n_1) = g(n_2) \\ \Downarrow \qquad f : \mathbb{N} \xrightarrow{\sim} X \wedge g : \mathbb{N} \xrightarrow{\sim} Y \\ m_1 = m_2 \wedge n_1 = n_2 \\ \Downarrow \\ (m_1, n_1) = (m_2, n_2) \end{gathered}$$

- **Surjectivity**: Suppose $(x, y) \in X \times Y$. Then

$$\begin{gathered} f : \mathbb{N} \xrightarrow{\sim} X \wedge g : \mathbb{N} \xrightarrow{\sim} Y \\ \Downarrow \\ \exists m, n \in \mathbb{N}, f(m) = x \wedge g(n) = y \\ \Downarrow \\ h(m, n) = (x, y) \end{gathered}$$

Therefore,

$$\begin{gathered} h : \mathbb{N} \times \mathbb{N} \xrightarrow{\sim} X \times Y \\ \Downarrow \qquad \text{Theorem 2.2.1 and lemma 2.3.13} \\ |X \times Y| = \aleph_0 \end{gathered}$$

□

Theorem 2.3.15.

$$\forall n \in \mathbb{N}, \forall X_1, X_2, \ldots, X_n, (\forall k \in [1,n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| = \aleph_0$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) : (\forall k \in [1,n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| = \aleph_0$$

Then

- **Base case:** Suppose
  - $|X_1| = \aleph_0$

  Then

$$\begin{gathered} |X_1| = \aleph_0 \\ \Downarrow \\ \left|\prod_{k=1}^{1} X_k\right| = \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Suppose $P(n)$ is true. Suppose
  - $\forall k \in [1,n+1]_{\mathbb{N}}, |X_k| = \aleph_0$

  Since $P(n)$ is true,

$$\left|\prod_{k=1}^{n} X_k\right| = \aleph_0$$

  Then

$$\begin{gathered} \prod_{k=1}^{n+1} X_k \sim \left(\prod_{k=1}^{n} X_k\right) \times X_{n+1} \\ \Downarrow \qquad \text{Theorem 2.3.14} \\ \left|\prod_{k=1}^{n+1} X_k\right| = \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n) \text{ is true}$$

□

Theorem 2.3.16.

$$|\mathbb{Q}| = \aleph_0$$

Proof. Define $f : \mathbb{Z} \times \mathbb{N} \twoheadrightarrow \mathbb{Q}$ by

$$\forall (m,n) \in \mathbb{Z} \times \mathbb{N}, f(m,n) = \frac{m}{n}$$

By theorems 2.3.5 and 2.3.14,

$$\begin{gathered} |\mathbb{Z} \times \mathbb{N}| = \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} \xrightarrow{\sim} \mathbb{Z} \times \mathbb{N} \end{gathered}$$

Also,

$$\begin{gathered} \mathbb{N} \subseteq \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| \geq \aleph_0 \end{gathered} \qquad \text{Corollary 2.2.6 and lemma 2.3.3}$$

By theorem 2.1.9,

$$\begin{gathered} f \circ g : \mathbb{N} \twoheadrightarrow \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.3.8}$$

$\square$

Lemma 2.3.17.

$$\left| \prod_{n=1}^{\infty} \{0,1\} \right| > \aleph_0$$

Proof. Define

$$\forall n, m \in \mathbb{N}, e_{nm} := \begin{cases} 1, & \text{if } m = n \\ 0, & \text{otherwise} \end{cases}$$

Define $f : \mathbb{N} \hookrightarrow \prod_{n=1}^{\infty} \{0,1\}$ by

$$\forall n \in \mathbb{N}, f(n) = (e_{nm})_{m=1}^{\infty}$$

By lemma 2.3.3 and theorem 2.3.2,

$$\left| \prod_{n=1}^{\infty} \{0,1\} \right| \geq \aleph_0$$

Assume $\left| \prod_{n=1}^{\infty} \{0,1\} \right| = \aleph_0$. Let

- $\prod_{n=1}^{\infty}\{0,1\} = \{a_1, a_2, a_3, \ldots\}$
- $\forall n \in \mathbb{N}, a_n = (a_{nm})_{m=1}^{\infty}$

Define

$$\forall m \in \mathbb{N}, b_m := 1 - a_{mm}$$

Then

$$\begin{gathered}\forall n \in \mathbb{N}, (b_m)_{m=1}^{\infty} \neq a_n \\ \Downarrow \\ (b_m)_{m=1}^{\infty} \notin \prod_{n=1}^{\infty}\{0,1\} \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\left|\prod_{n=1}^{\infty}\{0,1\}\right| > \aleph_0$$

$\square$

Theorem 2.3.18.

$$|\mathbb{R}| > \aleph_0$$

Proof. Consider

$$\begin{gathered}\mathbb{N} \subseteq \mathbb{R} \\ \Downarrow \\ |\mathbb{R}| \geq \aleph_0\end{gathered} \qquad \text{Lemma 2.3.3 and corollary 2.2.6}$$

Define $f : \mathbb{N} \hookrightarrow [0,1)$ by

$$\forall n \in \mathbb{N}, f(n) = \frac{1}{n+1}$$

By lemma 2.3.3 and theorem 2.3.2,

$$|[0,1)| \geq \aleph_0$$

Define $g : [0,1) \to \prod_{n=1}^{\infty}\{0,1\}$ by

$$\forall x \in [0,1), g(x) = \text{Greedy}_2(x)$$

By theorem 1.6.13,

$$g : [0,1) \hookrightarrow \prod_{n=1}^{\infty}\{0,1\}$$

Define

$$Y := g([0,1)) \subseteq \prod_{n=1}^{\infty} \{0,1\}$$

Then

$$g|^{Y} : [0,1) \xrightarrow{\sim} Y$$

Assume $|[0,1)| = \aleph_0$. By theorem 2.2.1,

$$|Y| = \aleph_0$$

Define

$$Z := \prod_{n=1}^{\infty} \{0,1\} \setminus Y$$

By Theorems 1.6.14 and 1.6.15,

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \,\middle|\, \exists N \in \mathbb{N}, \forall n \in [N,\infty)_{\mathbb{N}}, a_n = 1 \right\}$$

Define

$$\forall N \in \mathbb{N}, Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \,\middle|\, \forall n \in [N,\infty)_{\mathbb{N}}, a_n = 1 \right\} \neq \varnothing$$

Then

- If $N = 1$, then

$$Z_1 = \{(1)_{n=1}^{\infty}\}$$
$$\Downarrow$$
$$|Z_1| = 1 < \aleph_0$$

- Otherwise, define $F_N : \prod_{n=1}^{N-1} \{0,1\} \xrightarrow{\sim} Z_N$ as

$$\forall (a_n)_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0,1\}, F_N\Big((a_n)_{n=1}^{N-1}\Big) = (a_n)_{n=1}^{N-1} + (1)_{n=N}^{\infty}$$

  By theorem 2.2.17,

$$\forall N \in \mathbb{N}, \left| \prod_{n=1}^{N-1} \{0,1\} \right| < \aleph_0$$
$$\Downarrow \qquad \text{Theorem 2.2.8}$$
$$\forall N \in [2,\infty)_{\mathbb{N}}, |Z_N| < \aleph_0$$

Therefore,

$$Z = \bigcup_{N=1}^{\infty} Z_N$$
$$\Downarrow \qquad \text{Theorem 2.3.9}$$
$$|Z| \leq \aleph_0$$

By corollary 2.3.11,

$$\prod_{n=1}^{\infty} \{0,1\} = Y \cup Z$$
$$\Downarrow$$
$$\left|\prod_{n=1}^{\infty} \{0,1\}\right| = \aleph_0$$
$$\Downarrow \qquad \text{Lemma 2.3.17}$$
$$\perp$$

Therefore,

$$|[0,1)| > \aleph_0$$
$$\Downarrow$$
$$|\mathbb{R}| > \aleph_0$$

□

2.4 Metric Spaces

Definition 2.4.1. A function $d : X \times X \to \mathbb{R}$ is called a **metric** on X if for all $p, q, r \in X$,

- **(A)** $d(p,q) \geq 0$
- **(B)** $d(p,q) = 0 \iff p = q$
- **(C)** $d(p,q) = d(q,p)$
- **(D)** $d(p,r) \leq d(p,q) + d(q,r)$

(D) is called the **triangle inequality**. The pair (X,d) is called a **metric space**.

Theorem 2.4.2. *Suppose $n \in \mathbb{N}$. Define $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ as*

$$d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

*for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then d is a metric on $\mathbb{R}^n$. In particular, d is called the **standard metric** on $\mathbb{R}^n$.*

Proof. Suppose $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$.

- $d(\boldsymbol{x}, \boldsymbol{y}) \geq 0$ is trivial.
- By definition 1.9.3,

$$\begin{aligned} d(\boldsymbol{x}, \boldsymbol{y}) = 0 &\iff \|\boldsymbol{x} - \boldsymbol{y}\| = 0 \\ &\iff \forall k \in [1, n]_{\mathbb{N}}, x_k = y_k \\ &\iff \boldsymbol{x} = \boldsymbol{y} \end{aligned}$$

- $d(\boldsymbol{x}, \boldsymbol{y}) = d(\boldsymbol{y}, \boldsymbol{x})$ is trivial.
- By theorem 1.9.4,

$$\begin{aligned} d(\boldsymbol{x}, \boldsymbol{z}) &= \|\boldsymbol{x} - \boldsymbol{z}\| = \|(\boldsymbol{x} - \boldsymbol{y}) + (\boldsymbol{y} - \boldsymbol{z})\| \\ &\leq \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y} - \boldsymbol{z}\| = d(\boldsymbol{x}, \boldsymbol{y}) + d(\boldsymbol{y}, \boldsymbol{z}) \end{aligned}$$

Therefore, d is a metric on $\mathbb{R}^n$. □

Definition 2.4.3. Suppose $k \in \mathbb{N}$. A set $E \subseteq \mathbb{R}^k$ is called a k**-cell** if

$$E = \left\{ \boldsymbol{x} \in \mathbb{R}^k \,\middle|\, \forall j \in [1, k]_{\mathbb{N}}, a_j \leq x_j \leq b_j \right\}$$

where

$$\forall j \in [1, k]_{\mathbb{N}}, -\infty < a_j \leq b_j < +\infty$$

Definition 2.4.4. Suppose (X, d) is a metric space. Suppose $p \in X$ and $r \in \mathbb{R}^+$.

- The **neighborhood** of p with radius r is defined as

$$N_r(p) \coloneqq \{q \in X \mid d(p, q) < r\}$$

- The **deleted neighborhood** of p with radius r is defined as

$$\widetilde{N}_r(p) \coloneqq N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}$$

Definition 2.4.5. Suppose (X, d) is a metric space and $E \subseteq X$. Suppose $p \in X$.

- p is called an **limit point** of E if

$$\forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing$$

  The set of all limit points of E is denoted by E'.

- p is called a **isolated point** of E if

$$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

- p is called an **interior point** of E if

$$\exists r \in \mathbb{R}^+, N_r(p) \subseteq E$$

The set of all interior points of E is denoted by E°.

Definition 2.4.6. Suppose (X, d) is a metric space and $E \subseteq X$.

- E is said to be **open** if

$$\forall p \in E, p \in E^\circ$$

- E is said to be **closed** if

$$E' \subseteq E$$

Definition 2.4.7. Suppose (X, d) is a metric space and $E \subseteq X$. The **closure** $\overline{E}$ of E is defined as

$$\overline{E} := E \cup E'$$

Definition 2.4.8. Suppose (X, d) is a metric space and $E \subseteq X$.

- E is said to be **bounded** if

$$\exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)$$

- E is said to be **perfect** if

$$E = E'$$

- E is said to be **dense** if

$$\overline{E} = X$$

Theorem 2.4.9. *Suppose (X, d) is a metric space.*

$$\forall p \in X, \forall r \in \mathbb{R}^+, N_r(p) \text{ is open}$$

Proof. Suppose $q \in N_r(p)$. Then

$$d(p,q) < r$$
$$\Downarrow$$
$$\varepsilon := r - d(p,q) > 0$$

Suppose $s \in N_\varepsilon(q)$. Then

$$d(q,s) < \varepsilon$$
$$\Downarrow$$
$$d(p,s) \leq d(p,q) + d(q,s) < d(p,q) + \varepsilon = r$$
$$\Downarrow$$
$$s \in N_r(p)$$
$$\Downarrow$$
$$N_\varepsilon(q) \subseteq N_r(p)$$

Since q is arbitrary, $N_r(p)$ is open. □

Theorem 2.4.10. *Suppose* (X,d) *is a metric space and* $E \subseteq X$*.*

$$\forall p \in E', \forall r \in \mathbb{R}^+, |E \cap N_r(p)| \geq \aleph_0$$

Proof. Assume

$$\exists p \in E', \exists r \in \mathbb{R}^+, |E \cap N_r(p)| < \aleph_0$$

Define

$$S := (E \cap N_r(p)) \setminus \{p\} = E \cap \widetilde{N}_r(p) \subseteq E \cap N_r(p)$$

Then

$$p \in E'$$
$$\Downarrow$$
$$S \neq \varnothing$$

By theorem 2.2.5,

$$|E \cap N_r(p)| < \aleph_0$$
$$\Downarrow$$
$$|S| < \aleph_0$$

Define

$$T := \{d(p,q) \mid q \in S\} \neq \varnothing$$

Then

$$p \notin S$$
$$\Downarrow$$
$$0 \notin T$$

Define $f : S \twoheadrightarrow T$ as

$$f(q) = d(p, q)$$

By theorem 2.2.8,

$$|T| < \aleph_0$$

By theorem 2.2.9,

$$\exists \varepsilon \in \mathbb{R}^+, \varepsilon = \min T$$

Then

$$\begin{gathered} \varepsilon < r \\ \Downarrow \\ N_\varepsilon(p) \subseteq N_r(p) \end{gathered}$$

Also,

$$\begin{gathered} \forall q \in S, d(p, q) \geq \varepsilon \\ \Downarrow \\ E \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|E \cap N_r(p)| \geq \aleph_0$$

□

Theorem 2.4.11. *Suppose (X, d) is a metric space and $E \subseteq X$. Then*

$$E \textit{ is open} \iff E^c \textit{ is closed}$$

Proof. ($\implies$) Suppose E is open. Suppose $p \in (E^c)'$. Then

$$\forall r \in \mathbb{R}^+, E^c \cap \widetilde{N}_r(p) \neq \varnothing$$

If $p \in E$, then

$$\begin{gathered} p \in E \\ \Downarrow \\ \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(p) \subseteq E \\ \Downarrow \\ E^c \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} p \notin E \\ \Downarrow \\ p \in E^c \end{gathered}$$

Since p is arbitrary,

$$(E^c)' \subseteq E^c$$
$$\Downarrow$$
$$E^c \text{ is closed}$$

($\Longleftarrow$) Suppose E^c is closed. Suppose $p \in E$. If

$$\nexists r \in \mathbb{R}^+, N_r(p) \subseteq E,$$

then

$$\forall r \in \mathbb{R}^+, E^c \cap N_r(p) \neq \varnothing$$

Since $p \in E$,

$$p \notin E^c$$
$$\Downarrow$$
$$E^c \cap \widetilde{N}_r(p) \neq \varnothing$$
$$\Downarrow$$
$$p \in (E^c)'$$

Since E^c is closed,

$$p \in (E^c)'$$
$$\Downarrow$$
$$p \in E^c$$
$$\Downarrow$$
$$\perp$$

Therefore,

$$\exists r \in \mathbb{R}^+, N_r(p) \subseteq E$$
$$\Downarrow$$
$$p \in E^\circ$$

Since p is arbitrary, E is open. $\square$

Corollary 2.4.12. *Suppose (X, d) is a metric space and $E \subseteq X$. Then*

$$E \textit{ is closed} \iff E^c \textit{ is open}$$

Theorem 2.4.13. *Suppose (X, d) is a metric space.*

$$\forall Y, (\forall y \in Y, E_y \subseteq X \textit{ is open}) \implies \bigcup_{y \in Y} E_y \textit{ is open}$$

Proof. Suppose $p \in \bigcup_{y\in Y} E_y$. Then

$$\exists y_0 \in Y, p \in E_{y_0}$$

Since E_{y_0} is open,

$$\begin{gathered}\exists r \in \mathbb{R}^+, N_r(p) \subseteq E_{y_0} \\ \Downarrow \\ N_r(p) \subseteq E_{y_0} \subseteq \bigcup_{y\in Y} E_y \\ \Downarrow \\ p \in \left(\bigcup_{y\in Y} E_y\right)^\circ\end{gathered}$$

Since p is arbitrary, $\bigcup_{y\in Y} E_y$ is open. □

Corollary 2.4.14. *Suppose (X, d) is a metric space.*

$$\forall Y, (\forall y \in Y, E_y \subseteq X \textit{ is closed}) \implies \bigcap_{y\in Y} E_y \textit{ is closed}$$

Proof. By corollary 2.4.12,

$$\forall y \in Y, (E_y)^c \text{ is open}$$

By theorem 2.4.13,

$$\bigcup_{y\in Y} (E_y)^c \text{ is open}$$

By theorem 2.4.11,

$$\bigcap_{y\in Y} E_y = \left(\bigcup_{y\in Y} (E_y)^c\right)^c \text{ is closed}$$

□

Theorem 2.4.15. *Suppose (X, d) is a metric space.*

$$(\forall k \in [1, n]_{\mathbb{N}}, E_k \subseteq X \textit{ is open}) \implies \bigcap_{k=1}^{n} E_k \textit{ is open}$$

Proof. Suppose $p \in \bigcap_{k=1}^{n} E_k$. Then

$$\forall j \in [1, n]_{\mathbb{N}}, p \in E_j$$

Since each E_j is open,

$$\exists r_j \in \mathbb{R}^+, N_{r_j}(p) \subseteq E_j$$

for each $j \in [1, n]_{\mathbb{N}}$. Define

$$R \coloneqq \{r_j \mid j \in [1, n]_{\mathbb{N}}\} \subseteq \mathbb{R}^+$$

Define $f : [1, n]_{\mathbb{N}} \twoheadrightarrow R$ as

$$f(j) = r_j$$

By lemma 2.2.4 and theorem 2.2.8,

$$|R| < \aleph_0$$

By theorem 2.2.9,

$$\exists r \in \mathbb{R}^+, r = \min R$$

Then

$$\begin{gathered} \forall j \in [1, n]_{\mathbb{N}}, N_r(p) \subseteq N_{r_j}(p) \subseteq E_j \\ \Downarrow \\ N_r(p) \subseteq \bigcap_{k=1}^{n} E_k \\ \Downarrow \\ p \in \left(\bigcap_{k=1}^{n} E_k \right)^{\circ} \end{gathered}$$

Since p is arbitrary, $\bigcap_{k=1}^{n} E_k$ is open. $\square$

Corollary 2.4.16. *Suppose (X, d) is a metric space.*

$$(\forall k \in [1, n]_{\mathbb{N}}, E_k \subseteq X \textit{ is closed}) \implies \bigcup_{k=1}^{n} E_k \textit{ is closed}$$

Proof. By corollary 2.4.12,

$$\forall k \in [1, n]_{\mathbb{N}}, (E_k)^c \text{ is open}$$

By theorem 2.4.15,

$$\bigcap_{k=1}^{n}(E_k)^c \text{ is open}$$

By theorem 2.4.11,

$$\bigcup_{k=1}^{n} E_k = \left(\bigcap_{k=1}^{n}(E_k)^c\right)^c \text{ is closed}$$

□

Theorem 2.4.17. *Suppose (X, d) is a metric space and $E \subseteq X$. Then*

(A) *$\overline{E}$ is closed*

(B) *$E = \overline{E} \iff E$ is closed*

(C) *If $F \subseteq X$ satisfies*

- *$E \subseteq F$*
- *F is closed*

then $\overline{E} \subseteq F$

Proof.

(A) Suppose $p \in \left(\overline{E}\right)^c$. Then

$$\begin{gathered}
p \notin E \wedge p \notin E' \\
\Downarrow \\
\exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) = \varnothing \\
\Downarrow \\
E \cap N_r(p) = \varnothing \\
\Downarrow \\
N_r(p) \subseteq E^c
\end{gathered}$$

By theorem 2.4.9, $N_r(p)$ is open. Then

$$\begin{gathered}
\forall q \in N_r(p), \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(q) \subseteq N_r(p) \subseteq E^c \\
\Downarrow \\
E \cap N_\varepsilon(q) = \varnothing \\
\Downarrow \\
E \cap \widetilde{N}_\varepsilon(q) = \varnothing \\
\Downarrow \\
q \notin E'
\end{gathered}$$

Since q is arbitrary,

$$N_r(p) \subseteq \left(E'\right)^c$$
$$\Downarrow$$
$$N_r(p) \subseteq \left(\overline{E}\right)^c = E^c \cap \left(E'\right)^c$$
$$\Downarrow$$
$$p \in \left(\left(\overline{E}\right)^c\right)^\circ$$

Since p is arbitrary,

$$\left(\overline{E}\right)^c \text{ is open}$$

By corollary 2.4.12,

$$\overline{E} \text{ is closed}$$

(B) E is closed $\iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$

(C) Suppose $F \subseteq X$ satisfies

- $E \subseteq F$
- F is closed

Suppose $p \in E'$. Then,

$$\forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing$$
$$\Downarrow$$
$$F \cap \widetilde{N}_r(p) \neq \varnothing$$
$$\Downarrow$$
$$p \in F' \subseteq F$$

Since p is arbitrary,

$$E' \subseteq F' \subseteq F$$
$$\Downarrow$$
$$\overline{E} = E \cup E' \subseteq F$$

$\square$

Theorem 2.4.18. *Suppose $E \subseteq \mathbb{R}$ satisfies*

- $E \neq \varnothing$
- *E is bounded above*

then

$$\exists \sup E \in \overline{E}$$

Also,

$$E \text{ is closed} \implies \exists \sup E \in E$$

Proof. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

If $\alpha \in E$, then

$$\begin{gathered} E \subseteq \overline{E} \\ \Downarrow \\ \alpha \in \overline{E} \end{gathered}$$

Otherwise,

$$\begin{gathered} \forall x \in \mathbb{R}^+, \alpha - x \text{ is not an upper bound of } E \\ \Downarrow \\ \exists p \in E, \alpha - x < p < \alpha \\ \Downarrow \\ p \in \widetilde{N}_x(\alpha) \cap E \\ \Downarrow \\ \alpha \in E' \end{gathered}$$

Therefore,

$$\begin{gathered} E' \subseteq \overline{E} \\ \Downarrow \\ \alpha \in \overline{E} \end{gathered}$$

□

Theorem 2.4.19. *Suppose (X, d) is a metric space and $E \subseteq X$. Then*

$$\forall p \in X, p \in E \setminus E' \iff p \textit{ is an isolated point of } E$$

Proof. **($\implies$)** Suppose $p \in E \setminus E'$. Then

$$\begin{gathered} p \notin E' \\ \Downarrow \\ \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \\ (E \cap N_r(p)) \setminus \{p\} = \varnothing \end{gathered}$$

Therefore,

$$\begin{gathered} p \in E \wedge p \in N_r(p) \\ \Downarrow \\ E \cap N_r(p) = \{p\} \\ \Downarrow \\ p \text{ is an isolated point of } E \end{gathered}$$

($\impliedby$) Suppose p is an isolated point of E. Then

$$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

Therefore,

$$\begin{gathered} p \in E \\ \Downarrow \\ E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \\ p \notin E' \\ \Downarrow \\ p \in E \setminus E' \end{gathered}$$

□

Definition 2.4.20. Suppose (X, d) is a metric space and $E \subseteq X$. The **boundary** ∂E of E is defined as

$$\partial E \coloneqq \overline{E} \cap \overline{E^c}$$

Theorem 2.4.21. *Suppose (X, d) is a metric space and $E \subseteq X$. Then*

$$\partial E = \overline{E} \setminus E^\circ$$

Proof. We will show

$$\begin{gathered} (E^\circ)^c = \overline{E^c} \\ \Downarrow \\ \overline{E} \cap \overline{E^c} = \overline{E} \setminus E^\circ \end{gathered}$$

Suppose $p \in X$. Then

$$\begin{aligned} p \in (E^\circ)^c &\iff p \notin E^\circ \\ &\iff \forall r \in \mathbb{R}^+, N_r(p) \nsubseteq E \\ &\iff \forall r \in \mathbb{R}^+, N_r(p) \cap E^c \neq \varnothing \\ &\iff p \in \overline{E^c} \end{aligned}$$

□

Lemma 2.4.22. *Suppose $n \in \mathbb{N}$. Then*

$$\forall \boldsymbol{p} \in \mathbb{R}^n, \forall r \in \mathbb{R}^+, \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing$$

Proof. Suppose $\boldsymbol{p} \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$. Then

$$d\Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}, \boldsymbol{p}\Big) = \Big\|\Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}\Big) - \boldsymbol{p}\Big\| = \Big\|\frac{r}{2}\boldsymbol{e_1}\Big\| = \frac{r}{2}\|\boldsymbol{e_1}\| = \frac{r}{2} < r$$

where $\boldsymbol{e_1} = (1, 0, 0, \ldots, 0) \in \mathbb{R}^n$. Therefore,

$$\begin{gathered}\boldsymbol{p} \neq \left(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}\right) \in \widetilde{N}_r(\boldsymbol{p}) \\ \Downarrow \\ \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing\end{gathered}$$

□

Theorem 2.4.23. *Suppose $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then*

$$\partial E = \big(E \setminus E'\big) \cup \big(E^c \setminus (E^c)'\big) \cup \big(E' \cap (E^c)'\big)$$

This states that ∂E consists of

- *The set of all isolated points of E*
- *The set of all isolated points of E^c*
- *The set of all limit points of both E and E^c*

where the three sets are disjoint.

Proof. By definition,

$$\begin{aligned}\partial E &= \overline{E} \cap \overline{E^c} \\ &= \big(E \cup E'\big) \cap \big(E^c \cup (E^c)'\big) \\ &= (E \cap E^c) \cup \big(E \cap (E^c)'\big) \cup \big(E' \cap E^c\big) \cup \big(E' \cap (E^c)'\big)\end{aligned}$$

Consider

- $E = \big(E \setminus E'\big) \cup \big(E \cap E'\big)$
- $E^c = \big(E^c \setminus (E^c)'\big) \cup \big(E^c \cap (E^c)'\big)$

Then

- $E \cap (E^c)' = \big(\big(E \setminus E'\big) \cap (E^c)'\big) \cup \big(\big(E \cap E'\big) \cap (E^c)'\big)$
- $E' \cap E^c = \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap \big(E^c \cap (E^c)'\big)\big)$

Consider

- $\big(\big(E \cap E'\big) \cap (E^c)'\big) \subseteq E' \cap (E^c)'$
- $\big(E' \cap \big(E^c \cap (E^c)'\big)\big) \subseteq E' \cap (E^c)'$

Then

$$\begin{aligned}\partial E &= \varnothing \cup \big(\big(E \setminus E'\big) \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big) \\ &= \big(\big(E \setminus E'\big) \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big)\end{aligned}$$

Also,

$$\begin{gathered}\forall \boldsymbol{p} \in E \setminus E', \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(\boldsymbol{p}) = \varnothing \\ \Downarrow \\ \widetilde{N}_r(\boldsymbol{p}) \subseteq E^c\end{gathered}$$

By lemma 2.4.22,

$$\forall s \in \mathbb{R}^+, \widetilde{N}_s(\boldsymbol{p}) \cap \widetilde{N}_r(\boldsymbol{p}) = \widetilde{N}_{\min(s,r)}(\boldsymbol{p}) \neq \varnothing$$

Then

$$\begin{gathered}\forall s \in \mathbb{R}^+, \widetilde{N}_s(\boldsymbol{p}) \cap \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \widetilde{N}_s(\boldsymbol{p}) \cap \left(\widetilde{N}_r(\boldsymbol{p}) \cap E^c\right) \neq \varnothing \\ \Downarrow \\ \varnothing \neq \left(\widetilde{N}_s(\boldsymbol{p}) \cap E^c\right) \cap \widetilde{N}_r(\boldsymbol{p}) \subseteq \widetilde{N}_s(\boldsymbol{p}) \cap E^c \\ \Downarrow \\ E^c \cap \widetilde{N}_s(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \boldsymbol{p} \in (E^c)'\end{gathered}$$

Then

$$\begin{gathered}E \setminus E' \subseteq (E^c)' \\ \Downarrow \\ \big(E \setminus E'\big) \cap (E^c)' = E \setminus E'\end{gathered}$$

Similarly,

$$\begin{gathered}E^c \setminus (E^c)' \subseteq E' \\ \Downarrow \\ E' \cap \big(E^c \setminus (E^c)'\big) = E^c \setminus (E^c)'\end{gathered}$$

Therefore,

$$\partial E = \big(E \setminus E'\big) \cup \big(E^c \setminus (E^c)'\big) \cup \big(E' \cap (E^c)'\big)$$

□

Remark. Suppose (X, d) is a metric space and $Y \subseteq X$. Then

$$d|_{Y \times Y} : Y \times Y \to \mathbb{R}$$

satisfies the properties of a metric. Therefore, the pair

$$(Y, d|_{Y \times Y})$$

is also a metric space, which is called the **subspace** of (X, d). Since

- $d|_{Y \times Y}(p, q)$
- $(Y, d|_{Y \times Y})$

are cumbersome, we will write

- $d(p, q)$ instead of $d|_{Y \times Y}(p, q)$
- (Y, d) instead of $(Y, d|_{Y \times Y})$

for the sake of simplicity.

Definition 2.4.24. Suppose (X, d) is a metric space and $E \subseteq Y \subseteq X$. E is called **open relative** to Y if E is open in (Y, d). That is,

$$\forall p \in E, \exists r \in \mathbb{R}^+, Y \cap N_r(p) \subseteq E$$

Theorem 2.4.25. *Suppose (X, d) is a metric space and $E \subseteq Y \subseteq X$. If $V \subseteq X$ satisfies*

- *V is open in (X, d)*
- *$E = V \cap Y$*

then E is open relative to Y. The converse is also true.

Proof. ($\Longrightarrow$) Suppose $V \subseteq X$ satisfies

- V is open in (X, d)
- $E = V \cap Y$

Then

$$\begin{gathered}\forall p \in E, \exists r_p \in \mathbb{R}^+, N_{r_p}(p) \subseteq V \\ \Downarrow \\ Y \cap N_{r_p}(p) \subseteq V \cap Y = E\end{gathered}$$

($\Longleftarrow$) Suppose E is open relative to Y. Then

$$\forall p \in E, \exists r_p \in \mathbb{R}^+, Y \cap N_{r_p}(p) \subseteq E$$

Define

$$V := \bigcup_{p \in E} N_{r_p}(p)$$

By theorems 2.4.9 and 2.4.13, V is open in (X, d).

$$\begin{gathered}\forall p \in E, p \in N_{r_p}(p) \subseteq V \\ \Downarrow \\ E \subseteq Y \wedge E \subseteq V \\ \Downarrow \\ E \subseteq V \cap Y\end{gathered}$$

Also,

$$\begin{gathered}\forall p \in E, Y \cap N_{r_p}(p) \subseteq E \\ \Downarrow \\ \bigcup_{p \in E} \left(Y \cap N_{r_p}(p)\right) \subseteq E \\ \Downarrow \\ Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \\ \Downarrow \\ Y \cap V \subseteq E\end{gathered}$$

Therefore,

$$E = V \cap Y$$

□

2.5 Compact Sets

Definition 2.5.1. Suppose (X, d) is a metric space and $E \subseteq X$. The collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if

- $\forall \alpha \in A, G_\alpha$ is open
- $E \subseteq \bigcup_{\alpha \in A} G_\alpha$

A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of $\{G_\alpha\}_{\alpha \in A}$ of E if

- $S \subseteq A$
- $E \subseteq \bigcup_{\alpha \in S} G_\alpha$

E is called **compact** if every open cover of E has a finite subcover. That is,

$$\forall \{G_\alpha\}_{\alpha \in A}, \begin{bmatrix} \forall \alpha \in A, G_\alpha \text{ is open} \\ E \subseteq \bigcup_{\alpha \in A} G_\alpha \end{bmatrix} \implies \exists S \subseteq A, \begin{bmatrix} |S| < \aleph_0 \\ E \subseteq \bigcup_{\alpha \in S} G_\alpha \end{bmatrix}$$

Theorem 2.5.2. *Suppose (X, d) is a metric space and $E \subseteq Y \subseteq X$. Then*

$$E \text{ is compact relative to } Y \iff E \text{ is compact relative to } X$$

Proof. ($\implies$) Suppose E is compact relative to Y. Suppose $\{G_\alpha\}_{\alpha \in A}$ satisfies

- $\forall \alpha \in A, G_\alpha$ is open relative to X

- $E \subseteq \bigcup_{\alpha \in A} G_\alpha$

Define

$$H_\alpha = G_\alpha \cap Y$$

for each $\alpha \in A$. By theorem 2.4.25,

- $\forall \alpha \in A$, H_α is open relative to Y
- $E \subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha$

Since E is compact relative to Y,

$$\begin{gathered}
\{H_\alpha\}_{\alpha \in A} \text{ is an open cover of } E \text{ relative to } Y \\
\Downarrow \\
\exists S \subseteq A, \begin{bmatrix} |S| < \aleph_0 \\ E \subseteq \bigcup_{\alpha \in S} H_\alpha \end{bmatrix} \\
\Downarrow \\
E \subseteq \bigcup_{\alpha \in S} H_\alpha = \bigcup_{\alpha \in S} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S} G_\alpha \\
\Downarrow \\
\{G_\alpha\}_{\alpha \in S} \text{ is a finite subcover of } E \text{ relative to } X \\
\Downarrow \\
E \text{ is compact relative to } X
\end{gathered}$$

($\Longleftarrow$) Suppose E is a compact set relative to X. Suppose $\{H_\alpha\}_{\alpha \in A}$ satisfies

- $\forall \alpha \in A$, H_α is open relative to Y
- $E \subseteq \bigcup_{\alpha \in A} H_\alpha$

By theorem 2.4.25,

$$\begin{gathered}
\forall \alpha \in A, \exists G_\alpha, \begin{bmatrix} G_\alpha \text{ is open relative to } X \\ H_\alpha = G_\alpha \cap Y \end{bmatrix} \\
\Downarrow \\
E \subseteq \bigcup_{\alpha \in A} H_\alpha = \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha
\end{gathered}$$

Since E is compact relative to X,

$$\begin{gathered}
\{G_\alpha\}_{\alpha\in A} \text{ is an open cover of } E \text{ relative to } X \\
\Downarrow \\
\exists S \subseteq A, \begin{bmatrix} |S| < \aleph_0 \\ E \subseteq \bigcup_{\alpha\in S} G_\alpha \end{bmatrix} \\
\Downarrow \\
E \subseteq \left(\bigcup_{\alpha\in S} G_\alpha\right) \cap Y = \bigcup_{\alpha\in S} (G_\alpha \cap Y) = \bigcup_{\alpha\in S} H_\alpha \\
\Downarrow \\
\{H_\alpha\}_{\alpha\in S} \text{ is a finite subcover of } E \text{ relative to } Y \\
\Downarrow \\
E \text{ is compact relative to } Y
\end{gathered}$$

□

Remark. **Compactness** is preserved under the subspace topology, while **openness** and **closedness** are not, which is the reason why

"Suppose (X, d) is a metric space and $E \subseteq X$."

this kind of statement is always included in Section 2.4. In this textbook, you can assume that

- E is open $\implies$ E is open relative to X
- E is closed $\implies$ E is closed relative to X
- E is compact $\implies$ E is compact relative to X

if there is no specific mention of the subspace topology.

Theorem 2.5.3. *Suppose (X, d) is a metric space and $E \subseteq X$. Then*

$$E \textit{ is compact} \implies E \textit{ is closed}$$

Proof. Suppose E is compact. We will show

$$\begin{gathered}
E^c \text{ is open} \\
\Downarrow \\
E \text{ is closed}
\end{gathered}$$

Suppose $p \in E^c$. Define

$$r_q := \frac{d(p,q)}{2} > 0$$

for each $q \in E$. Then

$$\begin{gathered}\forall q \in E, q \in N_{r_q}(q) \\ \Downarrow \\ E \subseteq \bigcup_{q \in E} N_{r_q}(q) \\ \Downarrow \\ \left\{N_{r_q}(q)\right\}_{q \in E} \text{ is an open cover of } E\end{gathered}$$

Since E is compact,

$$\exists S \subseteq E, \begin{bmatrix} |S| < \aleph_0 \\ E \subseteq \bigcup_{q \in S} N_{r_q}(q) \end{bmatrix}$$

Define

$$T := \{r_q \mid q \in S\}$$

Define $f : S \twoheadrightarrow T$ as

$$f(q) = r_q$$

By theorem 2.2.8,

$$|T| < \aleph_0$$

By theorem 2.2.9,

$$\begin{gathered}\exists \varepsilon \in T, \varepsilon = \min T > 0 \\ \Downarrow \\ \forall x \in N_\varepsilon(p), \forall q \in S, d(x,p) < \varepsilon \leq r_q\end{gathered}$$

Assume

$$\exists q \in S, x \in N_{r_q}(q)$$

Then

$$\begin{gathered}x \in N_{r_q}(q) \\ \Downarrow \\ d(p,q) \leq d(p,x) + d(x,q) < r_q + r_q = 2r_q = d(p,q) \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

- $\forall q \in S, N_\varepsilon(p) \cap N_{r_q}(q) = \varnothing$
- $E \subseteq \bigcup_{q \in S} N_{r_q}(q)$

This implies

$$N_\varepsilon(p) \subseteq E^c$$

Since p is arbitrary, E^c is open. □

Theorem 2.5.4. *Suppose (X, d) is a metric space and $E \subseteq X$. Then*

$$E \text{ is compact} \implies E \text{ is bounded}$$

Proof. Suppose E is compact. If

$$E = \varnothing$$

then E is bounded. Otherwise, let $p \in E$. Define

$$G_n = N_n(p)$$

for each $n \in \mathbb{N}$. Then

$$\forall n, m \in \mathbb{N}, n \leq m \implies G_n \subseteq G_m$$

By theorem 1.6.2,

$$\begin{gathered} \forall q \in X, \exists n \in \mathbb{N}, d(p, q) < n \\ \Downarrow \\ E \subseteq X \subseteq \bigcup_{n=1}^{\infty} G_n \\ \Downarrow \\ \{G_n\}_{n=1}^{\infty} \text{ is an open cover of } E \end{gathered}$$

Since E is compact,

$$\exists S \subseteq \mathbb{N}, \begin{bmatrix} |S| < \aleph_0 \\ E \subseteq \bigcup_{n \in S} G_n \end{bmatrix}$$

By theorem 2.2.9,

$$\begin{gathered} |S| < \aleph_0 \\ \Downarrow \\ \exists m \in S, m = \max S \\ \Downarrow \\ E \subseteq \bigcup_{n \in S} G_n \subseteq G_m = N_m(p) \\ \Downarrow \\ E \text{ is bounded} \end{gathered}$$

□

Theorem 2.5.5. *Suppose (X, d) is a metric space and $F \subseteq E \subseteq X$. Then*

$$\begin{bmatrix} E \text{ is compact} \\ F \text{ is closed} \end{bmatrix} \implies F \text{ is compact}$$

Proof. Suppose

- E is compact
- F is closed

Suppose $\{H_\alpha\}_{\alpha \in A}$ is an open cover of F. Then

$$X = F \cup F^c \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c = \bigcup_{\alpha \in A} (H_\alpha \cup F^c)$$

By corollary 2.4.12 and theorem 2.4.13,

$$\begin{gathered} F^c \text{ is open} \\ \Downarrow \\ \forall \alpha \in A, H_\alpha \cup F^c \text{ is open} \\ \Downarrow \\ \{H_\alpha \cup F^c\}_{\alpha \in A} \text{ is an open cover of } E \end{gathered}$$

Since E is compact,

$$\exists B \subseteq A, \begin{bmatrix} |B| < \aleph_0 \\ E \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) \end{bmatrix}$$

Therefore,

$$\begin{gathered} F \subseteq E \\ \Downarrow \\ F \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) = \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c \\ \Downarrow \\ F \subseteq \bigcup_{\alpha \in B} H_\alpha \\ \Downarrow \\ F \text{ is compact} \end{gathered}$$

□

Theorem 2.5.6. *Suppose (X, d) is a metric space and $F \subseteq E \subseteq X$. Then*

$$\begin{bmatrix} E \text{ is compact} \\ |F| \geq \aleph_0 \end{bmatrix} \implies F' \cap E \neq \varnothing$$

Proof. Suppose that

- E is compact
- $|F| \geq \aleph_0$

Assume

$$F' \cap E = \varnothing$$

Then

$$\begin{gathered}\forall p \in E, \exists r_p \in \mathbb{R}^+, \widetilde{N}_{r_p}(p) \cap F = \varnothing \\ \Downarrow \\ N_{r_p}(p) \cap F \subseteq \{p\}\end{gathered}$$

Define

$$G_\alpha = N_{r_\alpha}(\alpha)$$

for each $\alpha \in E$. Then

$$\begin{gathered}\forall \alpha \in E, \alpha \in G_\alpha \\ \Downarrow \\ E \subseteq \bigcup_{\alpha \in E} G_\alpha \\ \Downarrow \\ \{G_\alpha\}_{\alpha \in E} \text{ is an open cover of } E\end{gathered}$$

Since E is compact, by theorem 2.2.5

$$\begin{gathered}\exists S \subseteq E, \begin{bmatrix} |S| < \aleph_0 \\ E \subseteq \bigcup_{\alpha \in S} G_\alpha \end{bmatrix} \\ \Downarrow \\ E \subseteq \bigcup_{\alpha \in S} G_\alpha \\ \Downarrow \\ F \subseteq E \subseteq \bigcup_{\alpha \in S} G_\alpha \\ \Downarrow \\ F = F \cap \bigcup_{\alpha \in S} G_\alpha = \bigcup_{\alpha \in S} (F \cap G_\alpha) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$F' \cap E \neq \varnothing$$

$\square$

Theorem 2.5.7. *Suppose* (X, d) *is a metric space. If*

$$\forall \alpha \in A, K_\alpha \subseteq X \textit{ is compact}$$

then

$$\left(\forall B \subseteq A, \begin{bmatrix} |B| < \aleph_0 \\ \bigcap_{\alpha \in B} K_\alpha \neq \varnothing \end{bmatrix}\right) \implies \bigcap_{\alpha \in A} K_\alpha \neq \varnothing$$

Proof. Suppose

- $\forall \alpha \in A, K_\alpha \subseteq X$ is compact
- $\forall B \subseteq A, \begin{bmatrix} |B| < \aleph_0 \\ \bigcap_{\alpha \in B} K_\alpha \neq \varnothing \end{bmatrix}$

Assume

$$\bigcap_{\alpha \in A} K_\alpha = \varnothing$$

Then

$$X = \left(\bigcap_{\alpha \in A} K_\alpha\right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c$$

By theorem 2.5.3 and corollary 2.4.12,

$$\begin{gathered} \forall \alpha \in A, K_\alpha \text{ is compact} \\ \Downarrow \\ \forall \alpha \in A, K_\alpha \text{ is closed} \\ \Downarrow \\ \forall \alpha \in A, (K_\alpha)^c \text{ is open} \\ \Downarrow \\ \{(K_\alpha)^c\}_{\alpha \in A} \text{ is an open cover of } X \end{gathered}$$

Then

$$\begin{gathered} K_\alpha \subseteq X \subseteq \bigcup_{\beta \in A} (K_\beta)^c \\ \Downarrow \\ \{(K_\beta)^c\}_{\beta \in A} \text{ is an open cover of } K_\alpha \end{gathered}$$

for all $\alpha \in A$. Since K_α is compact,

$$\begin{gathered}
\exists B_\alpha \subseteq A, \begin{bmatrix} |B_\alpha| < \aleph_0 \\ K_\alpha \subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c \end{bmatrix} \\
\Downarrow \\
K_\alpha \subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c = \left(\bigcap_{\beta \in B_\alpha} K_\beta \right)^c \\
\Downarrow \\
K_\alpha \cap \left(\bigcap_{\beta \in B_\alpha} K_\beta \right) = \varnothing \\
\Downarrow \\
\bigcap_{\beta \in B_\alpha \cup \{\alpha\}} K_\beta = \varnothing
\end{gathered}$$

for all $\alpha \in A$. By theorem 2.2.15,

$$\begin{gathered}
\begin{bmatrix} |B_\alpha| < \aleph_0 \\ |\{\alpha\}| < \aleph_0 \end{bmatrix} \\
\Downarrow \\
|B_\alpha \cup \{\alpha\}| < \aleph_0 \\
\Downarrow \\
\perp
\end{gathered}$$

Therefore,

$$\bigcap_{\alpha \in A} K_\alpha \neq \varnothing$$

□

Lemma 2.5.8. *Suppose*

- $(a_n)_{n=1}^\infty$ *in* $\mathbb{R}$
- $(b_n)_{n=1}^\infty$ *in* $\mathbb{R}$
- $(I_n)_{n=1}^\infty$ *in* $\mathcal{P}(\mathbb{R})$

such that

- $\forall n \in \mathbb{N}, a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$
- $\forall n \in \mathbb{N}, I_n = [a_n, b_n]$

Then

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing$$

Proof. By the *least upper bound property*,

$$\begin{gathered}\{a_n\}_{n=1}^{\infty} \text{ is bounded above} \\ \Downarrow \\ \exists \alpha \in \mathbb{R}, \alpha = \sup\{a_n\}_{n=1}^{\infty} \\ \Downarrow \\ \forall m \in \mathbb{N}, a_m \leq \alpha\end{gathered}$$

Suppose $m \in \mathbb{N}$. Then

- $\forall n \in \mathbb{N}, n \leq m \implies a_n \leq a_m \leq b_m$
- $\forall n \in \mathbb{N}, n > m \implies a_n \leq b_n \leq b_m$

Therefore,

$$\begin{gathered}\forall m \in \mathbb{N}, b_m \text{ is an upper bound of } \{a_n\}_{n=1}^{\infty} \\ \Downarrow \\ \forall m \in \mathbb{N}, a_m \leq \alpha \leq b_m \\ \Downarrow \\ \forall m \in \mathbb{N}, \alpha \in I_m \\ \Downarrow \\ \alpha \in \bigcap_{n=1}^{\infty} I_n\end{gathered}$$

□

Theorem 2.5.9. *Suppose* $k \in \mathbb{N}$. *Suppose*

- $\forall j \in [1, k]_{\mathbb{N}}, (a_{n,j})_{n=1}^{\infty}$ *in* $\mathbb{R}$
- $\forall j \in [1, k]_{\mathbb{N}}, (b_{n,j})_{n=1}^{\infty}$ *in* $\mathbb{R}$
- $(I_n)_{n=1}^{\infty}$ *in* $\mathcal{P}(\mathbb{R}^k)$

such that

- $\forall n \in \mathbb{N}, \forall j \in [1, k]_{\mathbb{N}}, a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$
- $\forall n \in \mathbb{N}, I_n = \left\{ \boldsymbol{x} \in \mathbb{R}^k \;\middle|\; \forall j \in [1, k]_{\mathbb{N}}, a_{n,j} \leq x_j \leq b_{n,j} \right\}$

Then

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing$$

Proof. Define

$$\forall n \in \mathbb{N}, \forall j \in [1,k]_{\mathbb{N}}, I_{n,j} = [a_{n,j}, b_{n,j}]$$

By lemma 2.5.8,

$$\begin{gathered}
\forall n \in \mathbb{N}, \forall j \in [1,k]_{\mathbb{N}}, a_{n,j} \le a_{n+1,j} \le b_{n+1,j} \le b_{n,j} \\
\Downarrow \\
\forall j \in [1,k]_{\mathbb{N}}, \bigcap_{n=1}^{\infty} I_{n,j} \neq \varnothing \\
\Downarrow \\
\forall j \in [1,k]_{\mathbb{N}}, \exists \alpha_j \in \bigcap_{n=1}^{\infty} I_{n,j} \\
\Downarrow \\
\forall n \in \mathbb{N}, \forall j \in [1,k]_{\mathbb{N}}, a_{n,j} \le \alpha_j \le b_{n,j}
\end{gathered}$$

Define

$$\boldsymbol{\alpha} := (\alpha_1, \alpha_2, \ldots, \alpha_k) \in \mathbb{R}^k$$

Then

$$\begin{gathered}
\forall n \in \mathbb{N}, \boldsymbol{\alpha} \in I_n \\
\Downarrow \\
\boldsymbol{\alpha} \in \bigcap_{n=1}^{\infty} I_n
\end{gathered}$$

□

Theorem 2.5.10. *Suppose $k \in \mathbb{N}$. Then every k-cell is compact in $\left(\mathbb{R}^k, \|\cdot\|\right)$.*

Proof. Suppose I is a k-cell in $\mathbb{R}^k$. Define

$$I_1 := I = \prod_{j=1}^{k} [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k$$

where

$$\forall j \in [1,k]_{\mathbb{N}}, -\infty < a_{1,j} \le b_{1,j} < +\infty$$

Define

$$\forall j \in [1,k]_{\mathbb{N}}, \forall m \in [1,2]_{\mathbb{N}}, I_{1,j,m} := \begin{cases} \left[a_{1,j}, \dfrac{a_{1,j}+b_{1,j}}{2}\right], & \text{if } m = 1 \\ \left[\dfrac{a_{1,j}+b_{1,j}}{2}, b_{1,j}\right], & \text{if } m = 2 \end{cases}$$

Define

$$S_k := \prod_{j=1}^{k} \{1, 2\} = \left\{ \boldsymbol{m} \in \mathbb{R}^k \;\middle|\; \forall j \in [1, k]_{\mathbb{N}}, m_j \in [1, 2]_{\mathbb{N}} \right\}$$

Then

$$I_1 = \bigcup_{\boldsymbol{m} \in S_k} \left(\prod_{j=1}^{k} I_{1,j,m_j} \right)$$

By theorem 2.2.17,

$$|S_k| < \aleph_0$$

Suppose $\{G_\alpha\}_{\alpha \in A}$ satisfies

- $\forall \alpha \in A, G_\alpha$ is open in $\left(\mathbb{R}^k, \|\cdot\|\right)$
- $I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha$

Assume

$$\nexists B \subseteq A, \begin{bmatrix} |B| < \aleph_0 \\ I_1 \subseteq \bigcup_{\alpha \in B} G_\alpha \end{bmatrix}$$

Then

$$\forall \boldsymbol{m} \in S_k, \prod_{j=1}^{k} I_{1,j,m_j} \subseteq I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha$$
$$\Downarrow$$
$$\{G_\alpha\}_{\alpha \in A} \text{ is an open cover of } \prod_{j=1}^{k} I_{1,j,m_j}$$

Assume

$$\forall \boldsymbol{m} \in S_k, \exists A_{\boldsymbol{m}} \subseteq A, \begin{bmatrix} |A_{\boldsymbol{m}}| < \aleph_0 \\ \prod_{j=1}^{k} I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{\boldsymbol{m}}} G_\alpha \end{bmatrix}$$

By theorem 2.4.13,

$$\bigcup_{\alpha \in A_{\boldsymbol{m}}} G_\alpha \text{ is open}$$

Then

$$I_1 \subseteq \bigcup_{\boldsymbol{m} \in S_k} \left(\bigcup_{\alpha \in A_{\boldsymbol{m}}} G_\alpha \right) = \bigcup_{\alpha \in A} G_\alpha$$

where

$$A := \bigcup_{\boldsymbol{m} \in S_k} A_{\boldsymbol{m}}$$

By theorem 2.2.15,

$$\begin{gathered} |A| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\exists \boldsymbol{m_1} \in S_k, \nexists A_{\boldsymbol{m_1}} \subseteq A, \left[\begin{gathered} |A_{\boldsymbol{m_1}}| < \aleph_0 \\ \prod_{j=1}^{k} I_{1,j,(m_1)_j} \subseteq \bigcup_{\alpha \in A_{\boldsymbol{m_1}}} G_\alpha \end{gathered} \right]$$

Define

$$I_2 := \prod_{j=1}^{k} [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k$$

where $a_{2,j}, b_{2,j} \in \mathbb{R}$ is defined as

- $a_{2,j} := \begin{cases} a_{1,j}, & \text{if } (m_1)_j = 1 \\ \dfrac{a_{1,j} + b_{1,j}}{2}, & \text{if } (m_1)_j = 2 \end{cases}$
- $b_{2,j} := \begin{cases} \dfrac{a_{1,j} + b_{1,j}}{2}, & \text{if } (m_1)_j = 1 \\ b_{1,j}, & \text{if } (m_1)_j = 2 \end{cases}$

for all $j \in [1, k]_{\mathbb{N}}$. Then

$$\begin{gathered} \forall j \in [1, k]_{\mathbb{N}}, [a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \subseteq [a_{1,j}, b_{1,j}] \\ \Downarrow \\ I_2 = \prod_{j=1}^{k} I_{1,j,(m_1)_j} \subseteq I_1 \end{gathered}$$

Define

$$I_{2,j,m} := \begin{cases} \left[a_{2,j}, \dfrac{a_{2,j} + b_{2,j}}{2} \right], & \text{if } m = 1 \\ \left[\dfrac{a_{2,j} + b_{2,j}}{2}, b_{2,j} \right], & \text{if } m = 2 \end{cases}$$

for all $j \in [1, k]_{\mathbb{N}}$ and $m \in [1, 2]_{\mathbb{N}}$. Then

$$I_2 = \bigcup_{\boldsymbol{m} \in S_k} \left(\prod_{j=1}^{k} I_{2,j,m_j} \right)$$

We have shown that

- $\{G_\alpha\}_{\alpha \in A}$ is an open cover of I_2

- I_2 has no finite subcover

Then

$$\forall \boldsymbol{m} \in S_k, \prod_{j=1}^{k} I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha \in A} G_\alpha$$
$$\Downarrow$$
$$\{G_\alpha\}_{\alpha \in A} \text{ is an open cover of } \prod_{j=1}^{k} I_{2,j,m_j}$$

$$\prod_{j=1}^{k} I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

Therefore,

$$\exists \boldsymbol{m_2} \in S_k, \prod_{j=1}^{k} I_{2,j,(m_2)_j} \text{ has no finite subcover}$$

By repeating this process, we can construct the following sequences:

- The sequence $(\boldsymbol{m}_n)_{n=1}^{\infty}$ in S_k.
- The sequences $(a_{n,j})_{n=1}^{\infty}$ and $(b_{n,j})_{n=1}^{\infty}$ in $\mathbb{R}$ for all $j = 1, 2, \ldots, k$.
- The sequence $(I_{n,j,m})_{n=1}^{\infty}$ of intervals in $\mathbb{R}$ for all $j = 1, 2, \ldots, k$ and $m = 1, 2$.
- The sequence $(I_n)_{n=1}^{\infty}$ of k-cells in $\mathbb{R}^k$.

We have the following properties:

- $I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$.
- I_n has no finite subcover for all $n \in \mathbb{N}$.
- $a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ for all $j = 1, 2, \ldots, k$ and $n \in \mathbb{N}$.
- $b_{n+1,j} - a_{n+1,j} = \frac{1}{2}(b_{n,j} - a_{n,j}) \implies b_{n,j} - a_{n,j} = \frac{1}{2^{n-1}}(b_{1,j} - a_{1,j})$ for all $j = 1, 2, \ldots, k$ and $n \in \mathbb{N}$.

Let $\delta := \sqrt{\sum_{j=1}^{k}(b_{1,j} - a_{1,j})^2}$. Then we have

$$\|\boldsymbol{x} - \boldsymbol{y}\| \leq \sqrt{\sum_{j=1}^{k}(b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}} \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in I_n \text{ and } n \in \mathbb{N}.$$

By theorem 2.5.9, we have $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$. Suppose $\boldsymbol{\beta} \in \bigcap_{n=1}^{\infty} I_n$. Then there exists $\alpha_0 \in A$ such that $\boldsymbol{\beta} \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$B_\varepsilon(\boldsymbol{\beta}) \subseteq G_{\alpha_0}.$$

By theorem 1.6.4, there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \frac{\varepsilon}{2\delta}$. Since $\boldsymbol{\beta} \in I_N$, we have

$$\|\boldsymbol{\beta} - \boldsymbol{\gamma}\| \leq \frac{\delta}{2^{N-1}} < \varepsilon \quad \text{for all } \boldsymbol{\gamma} \in I_N,$$

which implies that $I_N \subseteq B_\varepsilon(\boldsymbol{\beta}) \subseteq G_{\alpha_0}$. Therefore, I_N has a finite subcover $\{G_{\alpha_0}\}$, which is a contradiction. Thus I has a finite subcover, and I is compact. □

Corollary 2.5.11. *Suppose $-\infty < a \leq b < +\infty$. Then*

$$[a, b] \textit{ is compact in } (\mathbb{R}, \|\cdot\|)$$

Theorem 2.5.12. *Suppose $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$E \textit{ is compact} \iff E \textit{ is closed and bounded.}$$

*This theorem is called the **Heine-Borel Theorem**.*

Proof. ($\Longrightarrow$) Suppose E is compact. E is closed by theorem 2.5.3, and E is bounded by theorem 2.5.4.

($\Longleftarrow$) Suppose E is closed and bounded. Then there exists some $\boldsymbol{p} \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ such that $E \subseteq B_r(\boldsymbol{p})$. Let I be a n-cell in $\mathbb{R}^n$ defined as

$$I := \prod_{j=1}^{n} [p_j - r, p_j + r] \subseteq \mathbb{R}^n.$$

Then we have $E \subseteq B_r(\boldsymbol{p}) \subseteq I$. By theorem 2.5.10, I is compact. Since E is closed, by theorem 2.5.5, E is compact. □

2.6 Connected Sets

Definition 2.6.1. Suppose (X, d) is a metric space and $E \subseteq X$. E is called **connected** if there is no pair of sets $A, B \subseteq X$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.

- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Such pair of sets A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.6.2. *Suppose $E \subseteq \mathbb{R}$. Then E is connected if and only if*

$$a < x < b \implies x \in E \quad \textit{for all } a, b \in E \textit{ and } x \in \mathbb{R}.$$

Proof. ($\implies$) Suppose $E \subseteq \mathbb{R}$ is connected. Assume there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Define

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \varnothing$ and $B \neq \varnothing$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\impliedby$) Suppose $E \subseteq \mathbb{R}$ satisfies

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume E is disconnected. Then there exist separated sets $A, B \subseteq \mathbb{R}$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$. Without loss of generality, suppose that $a < b$. Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \varnothing$.
- $A \cap [a, b]$ is bounded above by b.

Then the *existence* of α is guaranteed by the *least upper bound property*. By theorem 2.4.18, we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By definition 2.4.5, it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \overline{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$.
- $\alpha \in \overline{A} \implies \alpha \notin B \implies \alpha \neq b$.

If $\alpha \notin A$, then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$ and $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \overline{B}.$$

Then there exists $r \in (0, b - \alpha)$ such that

$$\widetilde{N}_r(\alpha) \cap B = \varnothing \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that E is connected. $\square$

Chapter 3

Sequences and Series

3.1 Convergent Sequences

Definition 3.1.1. Suppose (X, d) is a metric space. A **sequence** $(x_n)_{n=1}^{\infty}$ is a function $f : \mathbb{N} \to X$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

A sequence $(x_n)_{n=1}^{\infty}$ is said to be **convergent** if there exists $x \in X$ such that

$$\forall \varepsilon \in \mathbb{R}^+, \exists N \in \mathbb{N}, \forall n \geq N, d(x_n, x) < \varepsilon$$

In this case, we say that

- $(x_n)_{n=1}^{\infty}$ **converges** to x.
- x is the **limit** of $(x_n)_{n=1}^{\infty}$.

We denote the limit of $(x_n)_{n=1}^{\infty}$ by

- $\lim\limits_{n\to\infty} x_n = x$.
- $x_n \to x$ as $n \to \infty$.

If $(x_n)_{n=1}^{\infty}$ is not convergent, it is said to be **divergent**.

Example 3.1.2. In $(\mathbb{R}, \|\cdot\|)$, define $(x_n)_{n=1}^{\infty}$ by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each $\varepsilon \in \mathbb{R}^+$, we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

Therefore, $(x_n)_{n=1}^{\infty}$ converges to 0.

In the subspace $\left(\mathbb{R}^+, \|\cdot\|\right)$ of $(\mathbb{R}, \|\cdot\|)$, define $(x_n)_{n=1}^{\infty}$ by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By theorem 1.6.2, for each $y \in \mathbb{R}^+$, there exists $N_y \in \mathbb{N}$ such that $N_y > \dfrac{1}{y}$. Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left|\frac{1}{n} - y\right| > \left|\frac{1}{N_y} - y\right| = \varepsilon_y.$$

Therefore, $(x_n)_{n=1}^{\infty}$ diverges in $\left(\mathbb{R}^+, \|\cdot\|\right)$.

Remark. Suppose (X, d) is a metric space. In this textbook, $(x_n)_{n=1}^{\infty}$ represents a sequence in (X, d), while $\{x_n\}_{n=1}^{\infty}$ represents a set defined as

$$\{x_n\}_{n=1}^{\infty} := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

Definition 3.1.3. Suppose (X, d) is a metric space. A sequence $(x_n)_{n=1}^{\infty}$ is said to be **bounded** if $\{x_n\}_{n=1}^{\infty}$ is bounded.

Theorem 3.1.4. *Suppose (X, d) is a metric space. Suppose $(x_n)_{n=1}^{\infty}$ is a convergent sequence. Then we have*

- ***(A)*** *$(x_n)_{n=1}^{\infty}$ is bounded.*
- ***(B)*** *The limit of $(x_n)_{n=1}^{\infty}$ is unique.*

Proof. Let x be the limit of $(x_n)_{n=1}^{\infty}$.

(A) Let $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < 1.$$

Define

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let $f : [1, N]_{\mathbb{N}} \to S$ defined as

$$f(k) = d(x_k, x) \quad \text{for all } k \in [1, N]_{\mathbb{N}}.$$

Then f is surjective. By lemma 2.2.7, S is finite. Then, by Theorem 2.2.9, S has a greatest element, say $M = \max S$. Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $(x_n)_{n=1}^{\infty}$ is bounded.

(B) Let y be the limit of $(x_n)_{n=1}^{\infty}$. Assume $x \neq y$. Then we have $r := d(x, y) > 0$. Let $\varepsilon = \dfrac{r}{2}$. Then we have

- There exists $N_x \in \mathbb{N}$ such that $n \geq N_x \implies d(x_n, x) < \dfrac{r}{2}$.
- There exists $N_y \in \mathbb{N}$ such that $n \geq N_y \implies d(x_n, y) < \dfrac{r}{2}$.

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x, y) \leq d(x, x_n) + d(x_n, y) < r,$$

which is a contradiction. Therefore, we have $x = y$.

$\square$

Theorem 3.1.5. *Suppose (X, d) is a metric space. Then the following statements hold:*

(A) *A sequence $(x_n)_{n=1}^{\infty}$ converges to $x \in X$ if and only if*

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) *Suppose $E \subseteq X$. If x is a limit point of E, then there exists $(x_n)_{n=1}^{\infty}$ in (X, d) such that*

$$\lim_{n \to \infty} x_n = x.$$

Proof.

(A) **($\implies$)** Suppose $(x_n)_{n=1}^{\infty}$ converge to $x \in X$. Then, for each $\varepsilon \in \mathbb{R}^+$, there exists $M_\varepsilon \in \mathbb{N}$ such that

$$n \geq M_\varepsilon \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_\varepsilon}^{\infty} \subseteq N_\varepsilon(x).$$

($\impliedby$) Suppose $(x_n)_{n=1}^{\infty}$ in (X, d) such that

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Suppose $E \subseteq X$ and x is a limit point of E. Then for every $\varepsilon \in \mathbb{R}^+$, we have $(N_\varepsilon(x) - \{x\}) \cap E \neq \varnothing$. Thus we can choose $(y_n)_{n=1}^{\infty}$ in (X, d) such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that $\lim_{n\to\infty} y_n = x$.

$\square$

3.2 Subsequences

3.3 Cauchy Sequences

3.4 Upper and Lower Limits

3.5 Some Special Sequences

3.6 Series

3.7 Series of Nonnegative Terms

3.8 The Number e

3.9 The Root and Ratio Tests

3.10 Power Series

3.11 Summation by Parts

3.12 Absolute Convergence

3.13 Addition and Multiplication of Series

3.14 Rearrangements

Chapter 4

Continuity

4.1 Limits of Functions

4.2 Continuous Functions

4.3 Continuity and Compactness

4.4 Continuity and Connectedness

4.5 Discontinuities

4.6 Monotonic Functions

4.7 Infinite Limits and Limits at Infinity

Chapter 5

Differentiation

Chapter 6

The Riemann-Stieltjes Integral

Chapter 7

Sequences and Series of Functions

7.1 Discussion of Main Problem

7.2 Uniform Convergence

7.3 Uniform Convergence and Continuity

7.4 Uniform Convergence and Integration

7.5 Uniform Convergence and Differentiation

7.6 Equicontinuous Families of Functions

7.7 Equicontinuous Families of Functions

7.8 The Stone-Weierstrass Theorem

Chapter 8

Some Special Functions

8.1 Power Series

8.2 The Exponential and Logarithmic Functions

8.3 The Trigonometric Functions

8.4 The Algebraic Completeness of the Complex Field

8.5 Fourier Series

8.6 The Gamma Function

Chapter 9

Functions of Several Variables

9.1 Linear Transformations

9.2 Differentiation

9.3 The Contraction Principle

9.4 The Inverse Function Theorem

9.5 The Implicit Function Theorem

9.6 The Rank Theorem

9.7 Determinants

9.8 Derivatives of Higher Order

9.9 Differentiation of Integrals

Chapter 10

Integration of Differential Forms

Chapter 11

The Lebesgue Theory

11.1 Set Functions

11.2 Construction of the Lebesgue Measure

11.3 Measure Spaces

11.4 Measurable Functions

11.5 Simple Functions

11.6 Integration

11.7 Comparison with the Riemann Integral

11.8 Integration of Complex Functions

11.9 Functions of Class $\mathscr{L}^2$

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